

Week 9: Logit and Probit

Samuel Rowe adapted from Wooldridge (2011), Long and Freese (2006), and Mitchell (2020)

2025-11-05

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```
bookdown::render_book()
```

Overview

We will cover the implementation of logit, probit, and multinomial logit.

1. Logit, Probit, and Linear Probability Model
2. Multinomial Logit
3. Ordinal Logit
4. Reshape

Chapter 1

Binary Outcomes

Married Women's Labor Force Participation

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use mroz.dta, clear
```

We'll use the data from Mroz (1987) to look at the probability of a married woman being in the labor force. Labor force participation is a binary response.

$$y = [0, 1]$$

We will estimate the coefficients of the linear probability model (LPM), the logit estimator, and the probit estimator. Then, we'll compare the marginal effects of all three estimators.

Summarize in the labor force

```
tabulate inlf
```

inlf	Freq.	Percent	Cum.
0	325	43.16	43.16
1	428	56.84	100.00
Total	753	100.00	

There are 325 women are not in the labor force and 428 women participating in the labor force. Our explanatory variables are non-wife income, education, experience, experience-squared, age, kids less than 6, kids greater than 6

$$y_i = \beta_0 + \beta_1 spouseinc_i + \beta_2 edu_i + \beta_3 exp_i + \beta_4 exp_i^2 + \beta_5 kidsLT6_i + \beta_6 kidsGT6_i + \varepsilon_i$$

```
est clear
```

```
eststo Logit: logit inlf nwifeinc educ exper expersq kidslt6 kidsge6
eststo Probit: probit inlf nwifeinc educ exper expersq kidslt6 kidsge6
esttab Logit Probit, mtitle
```

```
Iteration 0: log likelihood = -514.8732
Iteration 1: log likelihood = -422.78042
Iteration 2: log likelihood = -421.73851
Iteration 3: log likelihood = -421.73502
Iteration 4: log likelihood = -421.73502
```

Logistic regression	Number of obs	=	753
	LR chi2(6)	=	186.28
	Prob > chi2	=	0.0000
Log likelihood = -421.73502	Pseudo R2	=	0.1809

	inlf	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
nwifeinc		-.0301171	.0082431	-3.65	0.000	-.0462734 -.0139609
educ		.2520038	.0425492	5.92	0.000	.168609 .3353987
exper		.2057387	.0310518	6.63	0.000	.1448784 .266599
expersq		-.003913	.0009994	-3.92	0.000	-.0058718 -.0019541
kidslt6		-.9175126	.1742458	-5.27	0.000	-1.259028 -.5759971
kidsge6		.2226164	.0683456	3.26	0.001	.0886616 .3565713
_cons		-3.739707	.543217	-6.88	0.000	-4.804392 -2.675021

```
Iteration 0: log likelihood = -514.8732
Iteration 1: log likelihood = -422.36847
Iteration 2: log likelihood = -421.80202
Iteration 3: log likelihood = -421.80161
Iteration 4: log likelihood = -421.80161
```

Probit regression	Number of obs	=	753
	LR chi2(6)	=	186.14
	Prob > chi2	=	0.0000
Log likelihood = -421.80161	Pseudo R2	=	0.1808

inlf	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
nwifeinc	-.017188	.00474	-3.63	0.000	-.0264782	-.0078978
educ	.1501412	.02471	6.08	0.000	.1017105	.1985719
exper	.1240105	.0183233	6.77	0.000	.0880975	.1599236
expersq	-.0023694	.0005913	-4.01	0.000	-.0035284	-.0012103
kidslt6	-.5543317	.1038244	-5.34	0.000	-.7578238	-.3508395
kidsge6	.1307901	.0399186	3.28	0.001	.0525511	.2090292
_cons	-2.244553	.3146254	-7.13	0.000	-2.861207	-1.627899

	(1) Logit	(2) Probit
inlf		
nwifeinc	-0.0301*** (-3.65)	-0.0172*** (-3.63)
educ	0.252*** (5.92)	0.150*** (6.08)
exper	0.206*** (6.63)	0.124*** (6.77)
expersq	-0.00391*** (-3.92)	-0.00237*** (-4.01)
kidslt6	-0.918*** (-5.27)	-0.554*** (-5.34)
kidsge6	0.223** (3.26)	0.131** (3.28)
_cons	-3.740*** (-6.88)	-2.245*** (-7.13)
N	753	753

t statistics in parentheses

* p<0.05, ** p<0.01, *** p<0.001

We cannot compare the coefficients across the models. We will need to use marginal effects.

Average Marginal Effects (AME)

We will first look at average marginal effects. For average marginal effects, we estimate the marginal effects for each i and estimate an average.

$$AME = (\sum_{i=1}^n [g(\hat{\beta}_0 + x\hat{\beta})\beta_j] \Delta x_j) / n$$

Linear Probability Model (LPM)

Please note that we need the option *post* to start marginal effects.

```
est clear
quietly reg inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo LPM: margins, dydx(*) post
```

Logit

```
quietly logit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo Logit: margins, dydx(*) post
```

Probit

```
quietly probit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo Probit: margins, dydx(*) post
```

Compare

```
esttab LPM Logit Probit, mtitle
```

	(1) LPM	(2) Logit	(3) Probit
nwifeinc	-0.00341* (-2.35)	-0.00381* (-2.57)	-0.00362* (-2.51)
educ	0.0380*** (5.15)	0.0395*** (5.41)	0.0394*** (5.45)

exper	0.0395*** (6.96)	0.0368*** (7.14)	0.0371*** (7.20)
expersq	-0.000596** (-3.23)	-0.000563** (-3.18)	-0.000568** (-3.20)
age	-0.0161*** (-6.48)	-0.0157*** (-6.60)	-0.0159*** (-6.74)
kidslt6	-0.262*** (-7.81)	-0.258*** (-8.07)	-0.261*** (-8.20)
kidsge6	0.0130 (0.99)	0.0107 (0.81)	0.0108 (0.83)

N	753	753	753

t statistics in parentheses			
* p<0.05, ** p<0.01, *** p<0.001			

Marginal Effects at the Average (MEA)

For the marginal effects at the average, we set our x to their means within the scalar $g(\cdot)$

$$MEA = g(\hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \dots + \hat{\beta}_k \bar{x}_k) \beta_j \Delta x_j$$

Linear Probability Model LPM

```
est clear
quietly reg inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo LPM: margins, dydx(*) atmeans post
```

Logit

```
quietly logit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo Logit: margins, dydx(*) atmeans post
```

Probit

```
quietly probit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo Probit: margins, dydx(*) atmeans post
```

Compare

```
esttab LPM Logit Probit, mtitle
```

	(1) LPM	(2) Logit	(3) Probit
nwifeinc	-0.00341* (-2.35)	-0.00519* (-2.53)	-0.00470* (-2.48)
educ	0.0380*** (5.15)	0.0538*** (5.09)	0.0511*** (5.19)
exper	0.0395*** (6.96)	0.0501*** (6.40)	0.0482*** (6.57)
expersq	-0.000596** (-3.23)	-0.000767** (-3.10)	-0.000737** (-3.14)
age	-0.0161*** (-6.48)	-0.0214*** (-6.05)	-0.0206*** (-6.24)
kidslt6	-0.262*** (-7.81)	-0.351*** (-7.07)	-0.339*** (-7.32)
kidsge6	0.0130 (0.99)	0.0146 (0.80)	0.0141 (0.83)
N	753	753	753

t statistics in parentheses

* p<0.05, ** p<0.01, *** p<0.001

The analysis shows that the marginal effects are fairly close across the linear probability model, Logit model, and Probit model. One additional year of education increases the probability of being in the labor force by a range of 0.038 to 0.0395 or 3.8 to 3.95 percentage points. Interestingly, one additional

child less than six is associated with a drop in the probability of being in the labor force by a range of 0.258 to 0.262 or 25.8 to 26.2 percentage points.

Please not that around the means, our linear probability model, Logit, and Probit should be fairly similar. However, the marginal effects for the linear probability model are constant and will not vary across different values of x .

Odds Ratios

We can use the option, or to get odds ratios after running a logit.

$$OR = \frac{(OddsSuccess)}{(OddsFailure)} = \frac{p(1)/(1-p(1))}{p(0)/(1-p(0))}$$

```
logit inlf nwifeinc educ exper expersq age kidslt6 kidsge6, or
```

```
Iteration 0:  log likelihood =  -514.8732
Iteration 1:  log likelihood = -402.38502
Iteration 2:  log likelihood = -401.76569
Iteration 3:  log likelihood = -401.76515
Iteration 4:  log likelihood = -401.76515
```

Logistic regression	Number of obs	=	753
	LR chi2(7)	=	226.22
	Prob > chi2	=	0.0000
Log likelihood = -401.76515	Pseudo R2	=	0.2197

	inlf	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]
nwifeinc		.978881	.0082436	-2.53	0.011	.9628565 .9951723
educ		1.247536	.0541925	5.09	0.000	1.145717 1.358404
exper		1.228593	.0393849	6.42	0.000	1.153775 1.308263
expersq		.9968509	.0010129	-3.10	0.002	.9948676 .9988381
age		.9157386	.0133451	-6.04	0.000	.8899527 .9422715
kidslt6		.2361344	.0480734	-7.09	0.000	.158441 .3519257
kidsge6		1.061956	.0794234	0.80	0.422	.9171603 1.22961
_cons		1.530283	1.316609	0.49	0.621	.2834155 8.262655

One additional year of education is associated with a 1.25 times increase in the odds of being in the labor force (or an increase of 25%) holding all other variables constant. One additional child less than six decreases the odds of being in the labor force by a factor of 0.24 holding all other variables constant (or a decrease of 76%).

Marginal Effects of Education at different points along the curve

Linear Probability Model (LPM)

```
est clear
quietly reg inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo lpm: margins, at(educ=(0(2)20)) post
marginsplot, yline(0)
```

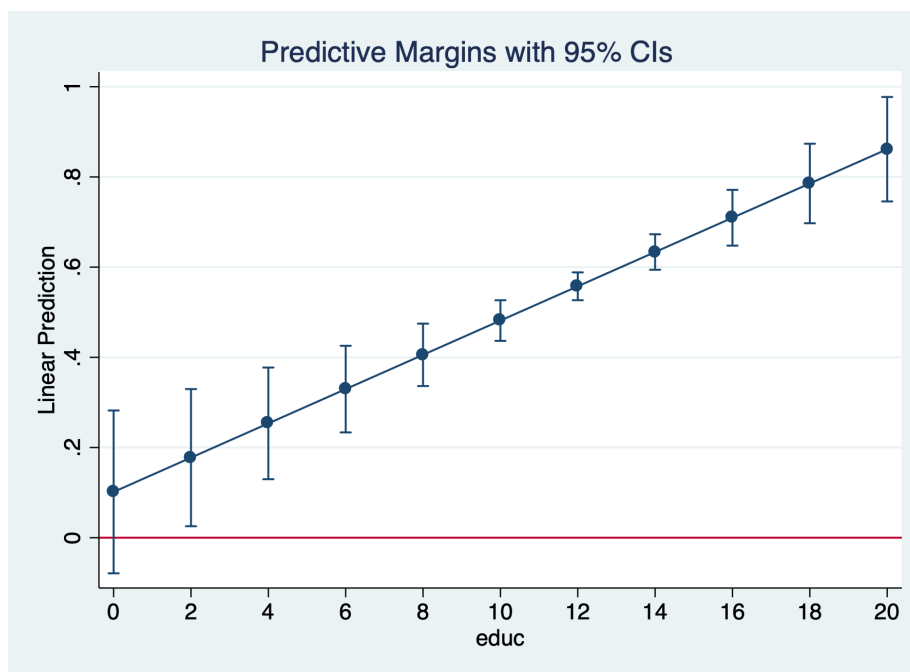


Figure 1.1: Marginal Effect of Education

Logit

```
quietly logit inlf nwifeinc educ exper expersq kidslt6 kidsge6
eststo logit1: margins, at(educ=(0(2)20)) post
marginsplot, yline(0)
```

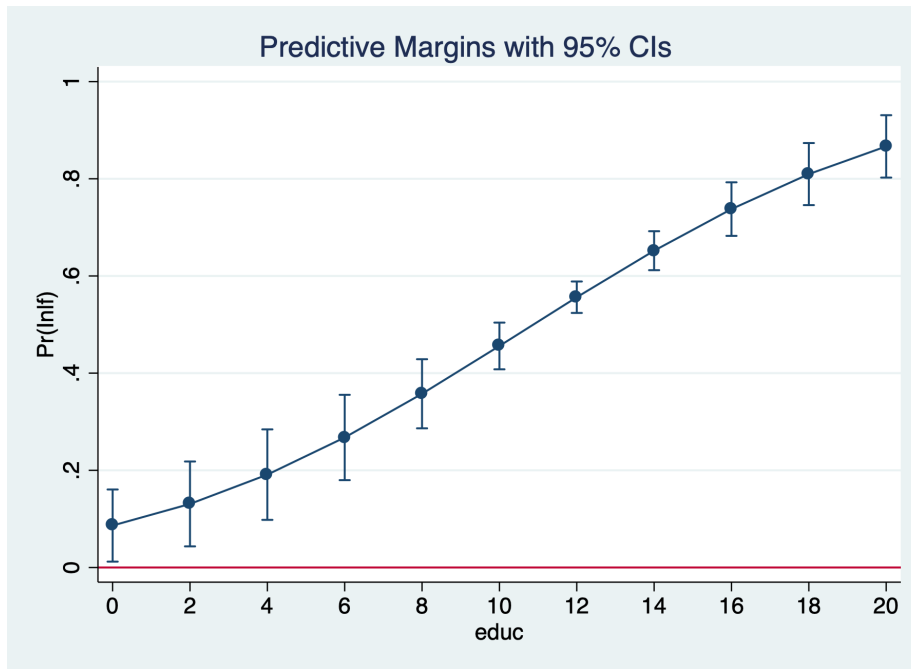


Figure 1.2: Marginal Effect of Education

Probit

```
quietly probit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo probit1: margins, at(educ=(0(2)20)) post
marginsplot, yline(0)
```

The predicted probability that a married women is in the labor force rises from 47.7% for 12 years of education to 71.2% for 16 years of education.

Coefficient Plot

We can also use `coefplot` to plot our data.

```
coefplot lpm logit1, at recast(line) ciopts(recast(rline) lpattern(dash))
coefplot lpm probit1, at recast(line) ciopts(recast(rline) lpattern(dash))
coefplot logit1 probit1, at recast(line) ciopts(recast(rline) lpattern(dash))
```

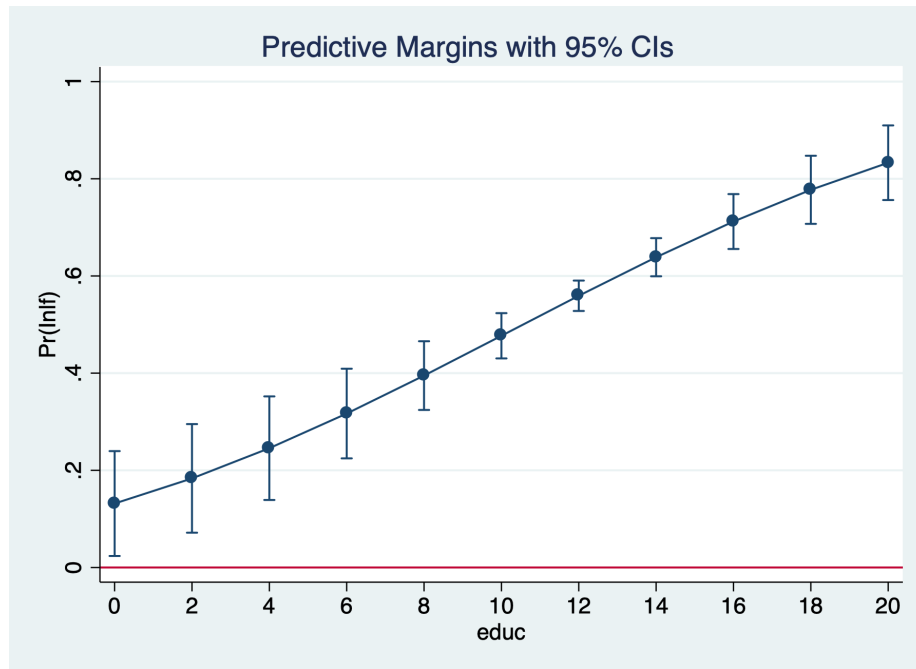


Figure 1.3: Marginal Effect of Education

Marginal Effects of Education at different points along the curve

Linear Probability Model (LPM)

```
quietly reg inlf nwifeinc educ exper expersq age kidslt6 kidsge6
margins, dydx(kidslt6) at(educ=(0(2)20))
marginsplot, yline(0)
```

Logit

```
quietly logit inlf nwifeinc educ exper expersq kidslt6 kidsge6
margins, dydx(kidslt6) at(educ=(0(2)20))
marginsplot, yline(0)
graph export "/Users/Sam/Desktop/Econ 645/Stata/week8_logitinlf.png", replace
```

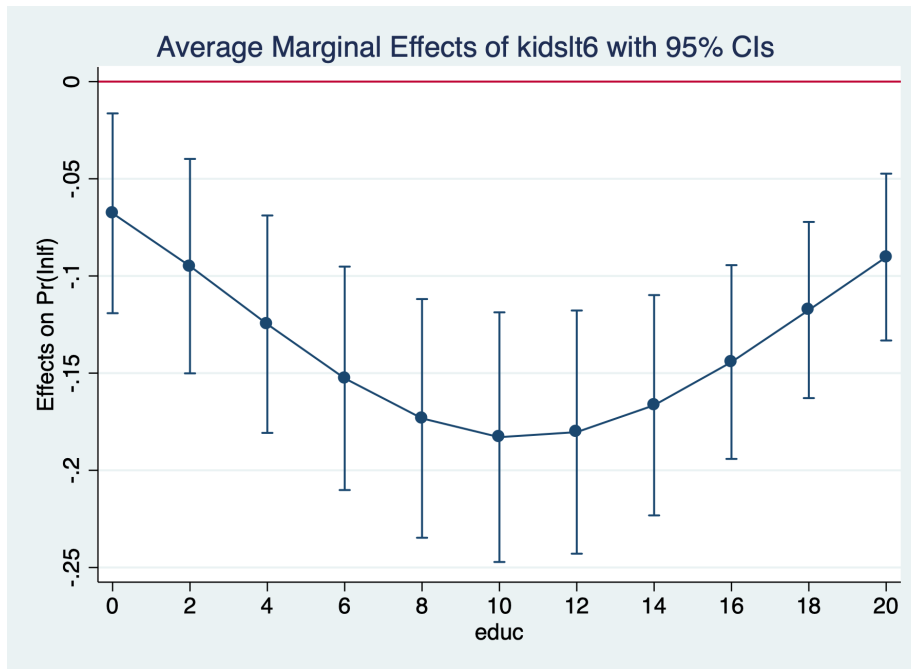



Figure 1.4: Marginal Effect of Kids Less Than 6

Probit

```
quietly probit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
margins, dydx(kidslt6) at(educ=(0(2)20))
marginsplot, yline(0)
```

The average marginal effect for an additional child less than 6 rises from -18.3 percentage points to -14.4 percentage points, but the difference does not appear to be statistically significant.

Chapter 2

Multinomial Logit

Adapted from Long and Freese (2006)

When we have nominal categorical variables, we cannot use a binary logit or probit. We will need to use multinomial logit or probit.

Lesson: Interpreting a nominal categorical dependent variable We will use Current Population Survey Data from September 2023 and 2024 to estimate the following model for labor force participation:

$$lfs_i = \beta_0 + \beta_1 edu_i + \beta_2 exper_i + \dots + u_i$$

.

There are three alternatives: Employed, Unemployed, and Not in the Labor Force.

2.1 Example Current Population Survey

```
use "/Users/Sam/Desktop/Econ 645/Data/CPS/mlogit_example.dta", clear
tab laborforce
tab laborforce, nolabel
```

Labor Force			
Status	Freq.	Percent	Cum.
-----+-----			
Employed	23,760	57.12	57.12
Unemployed	832	2.00	59.12
NILF	17,007	40.88	100.00

-----+-----			
Total		41,599	100.00
Labor Force			
Status		Freq.	Percent
			Cum.
-----+-----			
1		23,760	57.12
2		832	2.00
3		17,007	40.88
-----+-----			
Total		41,599	100.00

Our dependent variable has three nominal categories.

$$y = [1, 2, 3]$$

r

$$y = [Employed, Unemployed, NILF]$$

We use the *mlogit* command to run a multinomial logit to get log odds for $J - 1$ logits.

```
mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.union i.mar
```

```
Iteration 0: log likelihood = -31774.039
Iteration 1: log likelihood = -22975.759
Iteration 2: log likelihood = -22620.368
Iteration 3: log likelihood = -22573.244
Iteration 4: log likelihood = -22565.789
Iteration 5: log likelihood = -22564.343
Iteration 6: log likelihood = -22564.014
Iteration 7: log likelihood = -22563.941
Iteration 8: log likelihood = -22563.925
Iteration 9: log likelihood = -22563.922
Iteration 10: log likelihood = -22563.922
Iteration 11: log likelihood = -22563.922
Iteration 12: log likelihood = -22563.922
```

Multinomial logistic regression	Number of obs	=	41,599
	LR chi2(38)	=	18420.23
	Prob > chi2	=	0.0000
Log likelihood = -22563.922	Pseudo R2	=	0.2899

laborforcestatus	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
Employed	(base outcome)					
Unemployed						
educat						
HSD	-.2279524	.1119685	-2.04	0.042	-.4474066	-.0084982
Some College	-.4312492	.1303621	-3.31	0.001	-.6867542	-.1757442
AA/Vocational	-.6931682	.165487	-4.19	0.000	-1.017517	-.3688197
BS/BA	-.6706784	.1331892	-5.04	0.000	-.9317245	-.4096323
Graduate or Professional	-1.063383	.1745188	-6.09	0.000	-1.405433	-.7213323
exp	-.0291051	.0088759	-3.28	0.001	-.0465016	-.0117087
exp2	.0002524	.0001575	1.60	0.109	-.0000562	.0005611
race_ethnicity						
Asian/Pacific Islander	-.813795	.3012644	-2.70	0.007	-1.404262	-.2233275
Black	-.3951124	.2756281	-1.43	0.152	-.9353337	.1451088
Hispanic/Latino	-.6504449	.2691511	-2.42	0.016	-1.177971	-.1229184
White	-.7855686	.2616547	-3.00	0.003	-1.298402	-.2727347
Multiracial	-.4798449	.3366208	-1.43	0.154	-1.13961	.1799197
female						
Female	-.0004137	.0721979	-0.01	0.995	-.141919	.1410915
metroarea						
Nonmetro Area	-.1010417	.097797	-1.03	0.302	-.2927204	.0906369
Not Identified	.0635435	.346152	0.18	0.854	-.614902	.741989
union						
Union	-19.66515	2339.277	-0.01	0.993	-4604.564	4565.234
marital						
Divorced/Sep/Widowed	.6132556	.1175463	5.22	0.000	.3828691	.843642
Never Married	.7526873	.0988405	7.62	0.000	.5589634	.9464112
hryear4						
2024	.0228376	.071126	0.32	0.748	-.1165667	.1622419
_cons	-2.023219	.3012462	-6.72	0.000	-2.613651	-1.432787
NILF						
educat						
HSD	-.7898216	.0425444	-18.56	0.000	-.8732071	-.7064361
Some College	-.8941042	.0479568	-18.64	0.000	-.9880978	-.8001105
AA/Vocational	-1.143176	.0559087	-20.45	0.000	-1.252755	-1.033597

BS/BA		-1.493236	.0488335	-30.58	0.000	-1.588948	-1.33
Graduate or Professional		-1.746904	.0561914	-31.09	0.000	-1.857038	-1.63
exp		-.1490983	.003087	-48.30	0.000	-.1551487	-.14
exp2		.0031412	.0000469	66.92	0.000	.0030492	.003
race_ethnicity							
Asian/Pacific Islander		-.1995846	.1336975	-1.49	0.135	-.4616269	.062
Black		-.1053158	.1293382	-0.81	0.415	-.358814	.148
Hispanic/Latino		-.4785722	.1270357	-3.77	0.000	-.7275575	-.223
White		-.3378929	.1237786	-2.73	0.006	-.5804944	-.093
Multiracial		-.1802636	.1537778	-1.17	0.241	-.4816625	.123
female							
Female		.5781835	.0258942	22.33	0.000	.5274318	.623
metroarea							
Nonmetro Area		-.0069684	.0328081	-0.21	0.832	-.0712712	.051
Not Identified		-.0021328	.1282129	-0.02	0.987	-.2534254	.243
union							
Union		-20.25621	566.9973	-0.04	0.972	-1131.55	1093
marital							
Divorced/Sep/Widowed		-.0531382	.0373157	-1.42	0.154	-.1262757	.013
Never Married		.1464745	.0379587	3.86	0.000	.0720769	.220
hryear4							
2024		-.0398405	.0253899	-1.57	0.117	-.0896037	.003
_cons		1.072448	.1365064	7.86	0.000	.8049003	1.33

Note: 2044 observations completely determined. Standard errors questionable.

These are log-odds coefficients comparing the coefficients for unemployed and not in the labor force to employed. Let's use odds ratios.

2.1.1 Relative Risk Ratios

Akin to our odds ratios, relative risk ratios are

$$e^{\beta_j}$$

```
mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.union i.marital i.hry
```

```
Iteration 0: log likelihood = -31774.039
Iteration 1: log likelihood = -22975.759
Iteration 2: log likelihood = -22620.368
Iteration 3: log likelihood = -22573.244
Iteration 4: log likelihood = -22565.789
Iteration 5: log likelihood = -22564.343
Iteration 6: log likelihood = -22564.014
Iteration 7: log likelihood = -22563.941
Iteration 8: log likelihood = -22563.925
Iteration 9: log likelihood = -22563.922
Iteration 10: log likelihood = -22563.922
Iteration 11: log likelihood = -22563.922
Iteration 12: log likelihood = -22563.922
```

```
Multinomial logistic regression      Number of obs      =      41,599
                                      LR chi2(38)           =     18420.23
                                      Prob > chi2          =       0.0000
Log likelihood = -22563.922          Pseudo R2          =       0.2899
```

laborforcestatus		RRR	Std. Err.	z	P> z	[95% Conf. Interval]	
Employed		(base outcome)					
Unemployed							
	educat						
	HSD	.7961621	.0891451	-2.04	0.042	.6392839	.9915378
	Some College	.649697	.0846959	-3.31	0.001	.5032067	.8388326
	AA/Vocational	.4999895	.0827417	-4.19	0.000	.3614915	.6915501
	BS/BA	.5113616	.0681079	-5.04	0.000	.3938739	.6638943
Graduate or Professional		.3452858	.0602589	-6.09	0.000	.2452608	.4861042
	exp	.9713143	.0086213	-3.28	0.001	.954563	.9883596
	exp2	1.000252	.0001575	1.60	0.109	.9999438	1.000561
	race_ethnicity						
Asian/Pacific Islander		.443173	.1335123	-2.70	0.007	.2455481	.7998528
Black		.6736043	.1856643	-1.43	0.152	.3924549	1.156165
Hispanic/Latino		.5218136	.1404467	-2.42	0.016	.3079027	.8843359
White		.4558604	.119278	-3.00	0.003	.2729675	.7612947
Multiracial		.6188793	.2083277	-1.43	0.154	.3199439	1.197121
	female						

	Female	.9995863	.072168	-0.01	0.995	.8676915	1.0
	metroarea						
	Nonmetro Area	.9038953	.0883983	-1.03	0.302	.7462308	1.0
	Not Identified	1.065606	.3688616	0.18	0.854	.5406939	2.1
	union						
	Union	2.88e-09	6.74e-06	-0.01	0.993	0	
	marital						
	Divorced/Sep/Widowed	1.846433	.2170413	5.22	0.000	1.466486	2.3
	Never Married	2.122697	.2098085	7.62	0.000	1.748859	2.5
	hryear4						
	2024	1.0231	.072769	0.32	0.748	.8899707	1.1
	_cons	.1322292	.0398335	-6.72	0.000	.0732666	.23

NILF	educat						
	HSD	.4539258	.019312	-18.56	0.000	.4176101	.49
	Some College	.4089738	.0196131	-18.64	0.000	.3722842	.44
	AA/Vocational	.3188048	.017824	-20.45	0.000	.2857165	.3
	BS/BA	.2246446	.0109702	-30.58	0.000	.2041403	.24
	Graduate or Professional	.1743127	.0097949	-31.09	0.000	.1561345	.19
	exp	.8614844	.0026594	-48.30	0.000	.8562879	.86
	exp2	1.003146	.0000471	66.92	0.000	1.003054	1.0
	race_ethnicity						
	Asian/Pacific Islander	.8190709	.1095078	-1.49	0.135	.6302574	1.0
	Black	.9000402	.1164096	-0.81	0.415	.6985043	1.1
	Hispanic/Latino	.6196675	.0787199	-3.77	0.000	.4830875	.79
	White	.7132717	.0882878	-2.73	0.006	.5596216	.9
	Multiracial	.8350501	.1284122	-1.17	0.241	.6177555	1.1
	female						
	Female	1.782797	.046164	22.33	0.000	1.694575	1.8
	metroarea						
	Nonmetro Area	.9930558	.0325803	-0.21	0.832	.9312093	1.0
	Not Identified	.9978694	.1279397	-0.02	0.987	.7761376	1.2
	union						
	Union	1.60e-09	9.05e-07	-0.04	0.972	0	

marital							
Divorced/Sep/Widowed		.948249	.0353846	-1.42	0.154	.8813718	1.020201
Never Married		1.157745	.0439465	3.86	0.000	1.074738	1.247164
hryear4							
2024		.9609427	.0243982	-1.57	0.117	.9142935	1.009972
_cons		2.922525	.3989434	7.86	0.000	2.236473	3.819027

Note: 2044 observations completely determined. Standard errors questionable.

2.1.1.1 Unemployed relative to employed

For an individual with a high school degree relative to a high school drop out, the relative risk for being unemployed relative to employed decreases by a factor of 0.80

For an individual with some college relative to a high school drop out, the relative risk for being unemployed relative to employed decreases by a factor of 0.65

For an individual with an AA or vocational degree relative to a high school drop out, the relative risk for being unemployed relative to employed decreases by a factor of 0.50

For an individual with a BS/BA degree relative to a high school drop out, the relative risk for being unemployed relative to employed decreases by a factor of 0.51

For an individual with a graduate or professional degree relative to a high school drop out, the relative risk for being unemployed relative to employed decreases by a factor of 0.35

2.1.1.2 Not in the labor force relative to employed

For an individual with a high school degree relative to a high school drop out, the relative risk for being NILF relative to employed decreases by a factor of 0.45

For an individual with some college relative to a high school drop out, the relative risk for being NILF relative to employed decreases by a factor of 0.41

For an individual with an AA or vocational degree relative to a high school drop out, the relative risk for being NILF relative to employed decreases by a factor of 0.32

For an individual with a BS/BA degree relative to a high school drop out, the relative risk for being NILF relative to employed decreases by a factor of 0.22

For an individual with a graduate or professional degree relative to a high school drop out, the relative risk for being NILF relative to employed decreases by a factor of 0.17

2.2 Test for Independence of Irrelevant Alternative Assumption

Next, we need to test our Independence of Irrelevant Alternatives (IIA) assumption with a Hausman Test.

2.2.1 Estimate an Unrestricted model - We'll remove unemployed

```
quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.unemployed
estimates store unrestricted
```

We compare the log odds for unemployed and not in the labor force to being employed. Given that these are log odds, we'll need to convert them to Odds Ratios or find the marginal effects.

2.2.2 Estimate a Restricted Model to test Independence of Irrelevant Alternatives assumption

```
quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.unemployed
estimate store restricted
```

2.2.3 Use hausman command

```
hausman restricted unrestricted, alleq constant
```

Note: the rank of the differenced variance matrix (2) does not equal the number of coefficients. Make sure this is what you expect, or there may be problems computing the test. Examine the coefficients and standard errors for anything unexpected and possibly consider scaling your variables on a similar scale.

	---- Coefficients ----			
	(b)	(B)	(b-B)	sqrt(diag(V_b-V_B))

2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION 27

		restricted	unrestricted	Difference	S.E.

	educat				
	2	-.7860864	-.7898216	.0037352	.0035785
	3	-.89149	-.8941042	.0026142	.0032032
	4	-1.140012	-1.143176	.003164	.003669
	5	-1.486705	-1.493236	.0065306	.0036976
	6	-1.737118	-1.746904	.0097867	.0035815
	exp	-.1486206	-.1490983	.0004777	.0002218
	exp2	.0031335	.0031412	-7.78e-06	2.91e-06
	race_ethni-y				
	2	-.1712505	-.1995846	.0283341	.0120289
	3	-.0717454	-.1053158	.0335704	.0121412
	4	-.4427331	-.4785722	.0358391	.0115676
	5	-.3060009	-.3378929	.031892	.0117006
	6	-.1491388	-.1802636	.0311247	.0136923
	1.female	.5793374	.5781835	.0011539	.0018992
	metroarea				
	2	-.0070074	-.0069684	-.000039	.002025
	3	-.0064057	-.0021328	-.0042729	.0112617
	1.union	-20.20846	-20.25621	.0477553	.
	marital				
	2	-.0524486	-.0531382	.0006896	.0022895
	3	.1495841	.1464745	.0031096	.0023193
	_cons	1.00979	1.072448	-.0626578	.

b = consistent under Ho and Ha; obtained from mlogit
 B = inconsistent under Ha, efficient under Ho; obtained from mlogit

Test: Ho: difference in coefficients not systematic

chi2(2) = (b-B)'[(V_b-V_B)^(-1)](b-B)
 = 8.92
 Prob>chi2 = 0.0116
 (V_b-V_B is not positive definite)

Our chi-squared is 8.92, which means with reject the IIA assumption. We should consider a binary response here.

2.2.4 Predicted Probabilities

Next, we use Stata to estimate predicted probabilities

Adjusted predictions Number of obs = 41,599
Model VCE : OIM

	Delta-method					
	Margin	Std. Err.	z	P> z	[95% Conf. Interval]	
_cons	.7684672	4.830264	0.16	0.874	-8.698677	10.23561

Adjusted predictions Number of obs = 41,599
Model VCE : OIM

2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION29

```

Expression : Pr(laborforcestatus==Unemployed), predict(outcome(2))
at          : 1.educat      =    .1261088 (mean)
              2.educat      =    .286954 (mean)
              3.educat      =    .1488497 (mean)
              4.educat      =    .0969254 (mean)
              5.educat      =    .2079617 (mean)
              6.educat      =    .1332003 (mean)
              exp           =    32.89247 (mean)
              exp2          =   1467.595 (mean)
              1.race_eth~y  =    .0098079 (mean)
              2.race_eth~y  =    .0622611 (mean)
              3.race_eth~y  =    .0931032 (mean)
              4.race_eth~y  =    .1487776 (mean)
              5.race_eth~y  =    .6685738 (mean)
              6.race_eth~y  =    .0174764 (mean)
              0.female      =    .4804923 (mean)
              1.female      =    .5195077 (mean)
              1.metroarea   =    .8000673 (mean)
              2.metroarea   =    .1904132 (mean)
              3.metroarea   =    .0095195 (mean)
              0.union       =    .9508642 (mean)
              1.union       =    .0491358 (mean)
              1.marital     =    .5163586 (mean)
              2.marital     =    .1819515 (mean)
              3.marital     =    .3016899 (mean)

```

		Delta-method				
		Margin	Std. Err.	z	P> z	[95% Conf. Interval]

_cons		.0090578	1.03352	0.01	0.993	-2.016605 2.03472

```

Adjusted predictions          Number of obs      =    41,599
Model VCE      : OIM

```

```

Expression : Pr(laborforcestatus==NILF), predict(outcome(3))
at          : 1.educat      =    .1261088 (mean)
              2.educat      =    .286954 (mean)
              3.educat      =    .1488497 (mean)
              4.educat      =    .0969254 (mean)
              5.educat      =    .2079617 (mean)
              6.educat      =    .1332003 (mean)
              exp           =    32.89247 (mean)
              exp2          =   1467.595 (mean)

```

	Delta-method					
	Margin	Std. Err.	z	P> z	[95% Conf. Interval]	
_cons	.2224751	4.825216	0.05	0.963	-9.234774	9.679725

```
margins, dydx(*) predict(outcome(1))
margins, dydx(*) predict(outcome(2))
margins, dydx(*) predict(outcome(3))
```

	Delta-method				
	dy/dx	Std. Err.	z	P> z	[95% Conf. Inter
educat					

2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION³¹

	HSD	.1309012	.0072675	18.01	0.000	.1166572	.1451452
	Some College	.1499942	.0080928	18.53	0.000	.1341326	.1658558
	AA/Vocational	.1911979	.0091537	20.89	0.000	.1732569	.2091389
	BS/BA	.2407492	.0078942	30.50	0.000	.2252768	.2562215
Graduate or Professional		.2789487	.008527	32.71	0.000	.2622362	.2956613
	exp	.0219808	.0004288	51.26	0.000	.0211403	.0228213
	exp2	-.0004585	5.91e-06	-77.62	0.000	-.00047	-.0004469
	race_ethnicity						
Asian/Pacific Islander		.0417403	.0212777	1.96	0.050	.0000368	.0834438
Black		.0224391	.0206352	1.09	0.277	-.0180052	.0628834
Hispanic/Latino		.0802357	.0202145	3.97	0.000	.0406159	.1198555
White		.0618266	.0197636	3.13	0.002	.0230907	.1005625
Multiracial		.0348149	.024377	1.43	0.153	-.0129632	.082593
	female						
	Female	-.0839096	.0039186	-21.41	0.000	-.0915899	-.0762292
	metroarea						
	Nonmetro Area	.0022776	.0050425	0.45	0.652	-.0076056	.0121607
	Not Identified	-.0007629	.0197304	-0.04	0.969	-.0394337	.0379079
	union						
	Union	.4414329	.0020464	215.71	0.000	.4374219	.4454438
	marital						
	Divorced/Sep/Widowed	.0001626	.0057357	0.03	0.977	-.0110791	.0114043
	Never Married	-.030733	.005813	-5.29	0.000	-.0421263	-.0193397

Note: dy/dx for factor levels is the discrete change from the base level.

Average marginal effects
Model VCE : OIM

Number of obs = 41,599

```
Expression      : Pr(laborforcestatus==Unemployed), predict(outcome(2))
```

dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity 3.race_ethnicity
4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.female 2.metroarea 3.metroarea
3.marital

		Delta-method				
		dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]
	educat					

Note: dy/dx for factor levels is the discrete change from the base level.

```

Expression      : Pr(laborforcestatus==NILF), predict(outcome(3))
dy/dx w.r.t.    : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity 3.
                  4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.female 2.metroarea
                  3.marital

```

		Delta-method			
	dy/dx	Std. Err.	z	P> z	[95% Conf. Inter
educat					

2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION33

HSD		-.1330487	.0071954	-18.49	0.000	-.1471514	-.118946
Some College		-.1485456	.0079934	-18.58	0.000	-.1642124	-.1328789
AA/Vocational		-.1863908	.0090294	-20.64	0.000	-.2040881	-.1686934
BS/BA		-.2379236	.0077818	-30.57	0.000	-.2531757	-.2226716
Graduate or Professional		-.2708111	.0083689	-32.36	0.000	-.2872139	-.2544084
exp		-.0223855	.0004141	-54.06	0.000	-.0231971	-.0215739
exp2		.0004739	5.54e-06	85.61	0.000	.0004631	.0004848
race_ethnicity							
Asian/Pacific Islander		-.0240837	.0207242	-1.16	0.245	-.0647024	.0165349
Black		-.0124803	.0200649	-0.62	0.534	-.0518068	.0268463
Hispanic/Latino		-.0673952	.01964	-3.43	0.001	-.1058889	-.0289016
White		-.0454768	.0191993	-2.37	0.018	-.0831067	-.0078468
Multiracial		-.0235247	.0237295	-0.99	0.322	-.0700337	.0229843
female							
Female		.0876852	.0038386	22.84	0.000	.0801616	.0952088
metroarea							
Nonmetro Area		-.0004278	.0049292	-0.09	0.931	-.0100888	.0092332
Not Identified		-.0004879	.0192391	-0.03	0.980	-.0381959	.0372201
union							
Union		-.420348	.0020016	-210.01	0.000	-.424271	-.4164251
marital							
Divorced/Sep/Widowed		-.0112379	.0055186	-2.04	0.042	-.0220542	-.0004216
Never Married		.0178192	.0057407	3.10	0.002	.0065676	.0290707

Note: dy/dx for factor levels is the discrete change from the base level.

2.2.6 Marginal Effects at the Average

```
margins, dydx(*) atmeans predict(outcome(1))
margins, dydx(*) atmeans predict(outcome(2))
margins, dydx(*) atmeans predict(outcome(3))
```

Conditional marginal effects Number of obs = 41,599
Model VCE : OIM

Expression : Pr(laborforcestatus==Employed), predict(outcome(1))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity 3.race_ethn

	4.race_ethnicity	5.race_ethnicity	6.race_ethnicity	1.female	2.metroarea
at	3.marital				
	: 1.educat	=	.1261088 (mean)		
	2.educat	=	.286954 (mean)		
	3.educat	=	.1488497 (mean)		
	4.educat	=	.0969254 (mean)		
	5.educat	=	.2079617 (mean)		
	6.educat	=	.1332003 (mean)		
	exp	=	32.89247 (mean)		
	exp2	=	1467.595 (mean)		
	1.race_eth-y	=	.0098079 (mean)		
	2.race_eth-y	=	.0622611 (mean)		
	3.race_eth-y	=	.0931032 (mean)		
	4.race_eth-y	=	.1487776 (mean)		
	5.race_eth-y	=	.6685738 (mean)		
	6.race_eth-y	=	.0174764 (mean)		
	0.female	=	.4804923 (mean)		
	1.female	=	.5195077 (mean)		
	1.metroarea	=	.8000673 (mean)		
	2.metroarea	=	.1904132 (mean)		
	3.metroarea	=	.0095195 (mean)		
	0.union	=	.9508642 (mean)		
	1.union	=	.0491358 (mean)		
	1.marital	=	.5163586 (mean)		
	2.marital	=	.1819515 (mean)		
	3.marital	=	.3016899 (mean)		

		Delta-method					
		dy/dx	Std. Err.	z	P> z	[95% Conf. Inter	
	educat						
	HSD	.1755775	1.437386	0.12	0.903	-2.641647	2.99
	Some College	.196493	1.66501	0.12	0.906	-3.066867	3.46
	AA/Vocational	.240734	2.297249	0.10	0.917	-4.26179	4.74
	BS/BA	.2905963	3.204011	0.09	0.928	-5.98915	6.57
	Graduate or Professional	.3223929	3.782688	0.09	0.932	-7.09154	7.73
	exp	.0256925	.3928606	0.07	0.948	-.7443001	.799
	exp2	-.0005388	.0083342	-0.06	0.948	-.0168735	.016
	race_ethnicity						
	Asian/Pacific Islander	.0449909	.7906855	0.06	0.955	-1.504724	1.55
	Black	.0244574	.4314249	0.06	0.955	-.8211199	.870
	Hispanic/Latino	.0912758	1.237833	0.07	0.941	-2.334832	2.51
	White	.06928	.9827052	0.07	0.944	-1.856787	1.99


```

1.union      = .0491358 (mean)
1.marital    = .5163586 (mean)
2.marital    = .1819515 (mean)
3.marital    = .3016899 (mean)

```

		Delta-method					
		dy/dx	Std. Err.	z	P> z	[95% Conf. Inte	
	educat						
	HSD	.0005216	.0755305	0.01	0.994	-.1475154	.1475154
	Some College	-.0012432	.1546272	-0.01	0.994	-.3043069	.3043069
	AA/Vocational	-.0029475	.3436986	-0.01	0.993	-.6765844	.6765844
	BS/BA	-.0022892	.2747643	-0.01	0.993	-.5408173	.5391827
	Graduate or Professional	-.0047363	.5466531	-0.01	0.993	-1.076157	1.076157
	exp	.0000394	.0076697	0.01	0.996	-.014993	.014993
	exp2	-4.07e-06	.0004725	-0.01	0.993	-.0009301	.0009301
	race_ethnicity						
	Asian/Pacific Islander	-.0089668	1.008282	-0.01	0.993	-1.985164	1.985164
	Black	-.0051358	.5753938	-0.01	0.993	-1.132887	1.132887
	Hispanic/Latino	-.0069558	.7820821	-0.01	0.993	-1.539809	1.539809
	White	-.0084662	.9520543	-0.01	0.993	-1.874458	1.874458
	Multiracial	-.0058748	.6589403	-0.01	0.993	-1.297374	1.297374
	female						
	Female	-.0011625	.1324367	-0.01	0.993	-.2607337	.2607337
	metroarea						
	Nonmetro Area	-.0008667	.0980645	-0.01	0.993	-.1930696	.1930696
	Not Identified	.0005835	.0659971	0.01	0.993	-.1287684	.1287684
	union						
	Union	-.017075	.0009387	-18.19	0.000	-.0189148	-.0189148
	marital						
	Divorced/Sep/Widowed	.0055891	.631376	0.01	0.993	-1.231885	1.231885
	Never Married	.0067698	.7645739	0.01	0.993	-1.491768	1.491768

Note: dy/dx for factor levels is the discrete change from the base level.

```

Conditional marginal effects      Number of obs      =      41,599
Model VCE      : OIM

```

2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION37

```

Expression : Pr(laborforcestatus==NILF), predict(outcome(3))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity 3.race_ethnicity
               4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.female 2.metroarea 3.metroarea
               3.marital
at          : 1.educat      = .1261088 (mean)
              2.educat      = .286954 (mean)
              3.educat      = .1488497 (mean)
              4.educat      = .0969254 (mean)
              5.educat      = .2079617 (mean)
              6.educat      = .1332003 (mean)
              exp           = 32.89247 (mean)
              exp2          = 1467.595 (mean)
              1.race_eth~y  = .0098079 (mean)
              2.race_eth~y  = .0622611 (mean)
              3.race_eth~y  = .0931032 (mean)
              4.race_eth~y  = .1487776 (mean)
              5.race_eth~y  = .6685738 (mean)
              6.race_eth~y  = .0174764 (mean)
              0.female      = .4804923 (mean)
              1.female      = .5195077 (mean)
              1.metroarea   = .8000673 (mean)
              2.metroarea   = .1904132 (mean)
              3.metroarea   = .0095195 (mean)
              0.union       = .9508642 (mean)
              1.union       = .0491358 (mean)
              1.marital     = .5163586 (mean)
              2.marital     = .1819515 (mean)
              3.marital     = .3016899 (mean)

```

		Delta-method					
		dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
educat							
	HSD	-.1760991	1.475242	-0.12	0.905	-3.06752	2.715321
	Some College	-.1952498	1.746298	-0.11	0.911	-3.617932	3.227432
	AA/Vocational	-.2377865	2.409215	-0.10	0.921	-4.959762	4.484189
	BS/BA	-.2883071	3.317729	-0.09	0.931	-6.790936	6.214322
	Graduate or Professional	-.3176566	3.916079	-0.08	0.935	-7.993031	7.357718
exp							
	exp	-.0257319	.3987801	-0.06	0.949	-.8073266	.7558628
	exp2	.0005428	.0084077	0.06	0.949	-.0159359	.0170216
race_ethnicity							
	Asian/Pacific Islander	-.0360241	.5758254	-0.06	0.950	-1.164621	1.092573
	Black	-.0193216	.3135138	-0.06	0.951	-.6337973	.5951541

Hispanic/Latino		-.08432	1.269837	-0.07	0.947	-2.573155	2.40
White		-.0608138	.9093883	-0.07	0.947	-1.843182	1.72
Multiracial		-.0333911	.49753	-0.07	0.946	-1.008532	.94
female							
Female		.0993401	1.526768	0.07	0.948	-2.89307	3.09
metroarea							
Nonmetro Area		-.001017	.0285739	-0.04	0.972	-.0570207	.05
Not Identified		-.0002091	.0267367	-0.01	0.994	-.052612	.05
union							
Union		-.431732	.0031813	-135.71	0.000	-.4379673	-.42
marital							
Divorced/Sep/Widowed		-.0100452	.2043758	-0.05	0.961	-.4106144	.39
Never Married		.0242282	.419673	0.06	0.954	-.7983157	.84

 Note: dy/dx for factor levels is the discrete change from the base level.

2.2.7 Change the Base Alternative

We can also change the base or reference alternative with the `base()` option

```
mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.union i.marital
```

```
Iteration 0: log likelihood = -31774.039
Iteration 1: log likelihood = -22976.886
Iteration 2: log likelihood = -22621.724
Iteration 3: log likelihood = -22574.602
Iteration 4: log likelihood = -22567.151
Iteration 5: log likelihood = -22565.706
Iteration 6: log likelihood = -22565.376
Iteration 7: log likelihood = -22565.303
Iteration 8: log likelihood = -22565.287
Iteration 9: log likelihood = -22565.285
Iteration 10: log likelihood = -22565.284
Iteration 11: log likelihood = -22565.284
Iteration 12: log likelihood = -22565.284
```

Multinomial logistic regression	Number of obs	=	41,599
	LR chi2(36)	=	18417.51
	Prob > chi2	=	0.0000
Log likelihood = -22565.284	Pseudo R2	=	0.2898

2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION39

laborforcestatus		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
Employed							
	educat						
	HSD	.7894788	.0425405	18.56	0.000	.7061009	.8728567
	Some College	.8934956	.0479536	18.63	0.000	.7995083	.9874829
	AA/Vocational	1.142766	.055907	20.44	0.000	1.03319	1.252341
	BS/BA	1.492912	.0488292	30.57	0.000	1.397209	1.588616
	Graduate or Professional	1.746553	.056187	31.08	0.000	1.636428	1.856677
	exp	.1490952	.0030868	48.30	0.000	.1430452	.1551451
	exp2	-.0031411	.0000469	-66.92	0.000	-.0032331	-.0030491
	race_ethnicity						
	Asian/Pacific Islander	.199411	.1336958	1.49	0.136	-.062628	.46145
	Black	.1055321	.1293351	0.82	0.415	-.14796	.3590242
	Hispanic/Latino	.4793179	.1270317	3.77	0.000	.2303404	.7282954
	White	.3381891	.123776	2.73	0.006	.0955926	.5807856
	Multiracial	.1810197	.1537766	1.18	0.239	-.1203769	.4824163
	female						
	Female	-.5781724	.0258932	-22.33	0.000	-.6289222	-.5274227
	metroarea						
	Nonmetro Area	.007027	.0328055	0.21	0.830	-.0572707	.0713247
	Not Identified	.0004518	.1282484	0.00	0.997	-.2509105	.2518142
	union						
	Union	20.25617	567.0506	0.04	0.972	-1091.143	1131.655
	marital						
	Divorced/Sep/Widowed	.0530922	.0373158	1.42	0.155	-.0200455	.12623
	Never Married	-.1465344	.0379554	-3.86	0.000	-.2209255	-.0721432
	_cons	-1.052655	.1359142	-7.74	0.000	-1.319042	-.7862681
Unemployed							
	educat						
	HSD	.56128	.1124074	4.99	0.000	.3409657	.7815944
	Some College	.4618071	.1313983	3.51	0.000	.2042712	.719343
	AA/Vocational	.4492192	.1680784	2.67	0.008	.1197916	.7786467
	BS/BA	.8221872	.1354557	6.07	0.000	.5566988	1.087676
	Graduate or Professional	.6830694	.1777718	3.84	0.000	.334643	1.031496

	exp		.1200086	.0089721	13.38	0.000	.1024235	.13
	exp2		-.0028889	.0001584	-18.24	0.000	-.0031994	-.00
	race_ethnicity							
Asian/Pacific Islander			-.6146734	.3071598	-2.00	0.045	-1.216696	-.01
Black			-.2893004	.2809389	-1.03	0.303	-.8399305	.26
Hispanic/Latino			-.1705519	.2744134	-0.62	0.534	-.7083923	.36
White			-.4471628	.2665893	-1.68	0.093	-.9696683	.07
Multiracial			-.2985012	.342879	-0.87	0.384	-.9705317	.37
	female							
Female			-.5787555	.0737225	-7.85	0.000	-.7232489	-.4
	metroarea							
Nonmetro Area			-.0940971	.0993931	-0.95	0.344	-.2889041	.10
Not Identified			.0622817	.3526626	0.18	0.860	-.6289243	.75
	union							
Union			.5900779	2407.697	0.00	1.000	-4718.409	471
	marital							
Divorced/Sep/Widowed			.6663185	.1197798	5.56	0.000	.4315545	.90
Never Married			.6061848	.1022777	5.93	0.000	.4057242	.80
	_cons		-3.064479	.303667	-10.09	0.000	-3.659655	-2.4
-----+-----								
NILF			(base outcome)					

Note: 2044 observations completely determined. Standard errors questionable.

2.2.8 Marginal Effects and Marginsplot

Next, we will estimate and graph the average marginal effects for education and experience

```
quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.un
```

Education

For Employed

```
margins, dydx(educat) predict(outcome(1))
quietly marginsplot, allsimplelabels horizontal recast(scatter) name(Employed) yscale(
```


2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION⁴¹

Average marginal effects	Number of obs	=	41,599
Model VCE : OIM			

```
Expression : Pr(laborforcestatus==Employed), predict(outcome(1))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat
```

		Delta-method					
		dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
educat							
HSD		.1309012	.0072675	18.01	0.000	.1166572	.1451452
Some College		.1499942	.0080928	18.53	0.000	.1341326	.1658558
AA/Vocational		.1911979	.0091537	20.89	0.000	.1732569	.2091389
BS/BA		.2407492	.0078942	30.50	0.000	.2252768	.2562215
Graduate or Professional		.2789487	.008527	32.71	0.000	.2622362	.2956613

Note: dy/dx for factor levels is the discrete change from the base level.

Individuals with high school degree have 13.1 percentage point more likely to be employed compared to high school dropouts. Individuals with some college are 15 percentage points more likely to be employed compared to high school dropouts. Individuals with Associates or Vocational degrees are 19.1 percentage points more likely to be employed compared to high school dropouts. Individuals with a Bachelor's degree are 24.1 percentage points more likely to be employed, while individuals with a graduate degree are 27.9 percentage points more likely to be employed compared to high school dropouts.

For Unemployed

```
margins, dydx(educat) predict(outcome(2))
quietly marginsplot, allsimplelabels horizontal recast(scatter) name(Unemployed) yscale(reverse)
```

For Not in the Labor Force

```
margins, dydx(educat) predict(outcome(3))
quietly marginsplot, allsimplelabels horizontal recast(scatter) name(NILF) yscale(reverse) ytitle(
```

Combine graphs

```
graph combine Employed Unemployed NILF, ycommon title("AME of Education") ///
note("Source: Current Population Survey; 2 is HSD, 3 is Some College, 4 is AA, 5 is BA/BS, and 6
graph export "/Users/Sam/Desktop/Econ 645/Stata/week8_mnlmeducation.png", replace
```

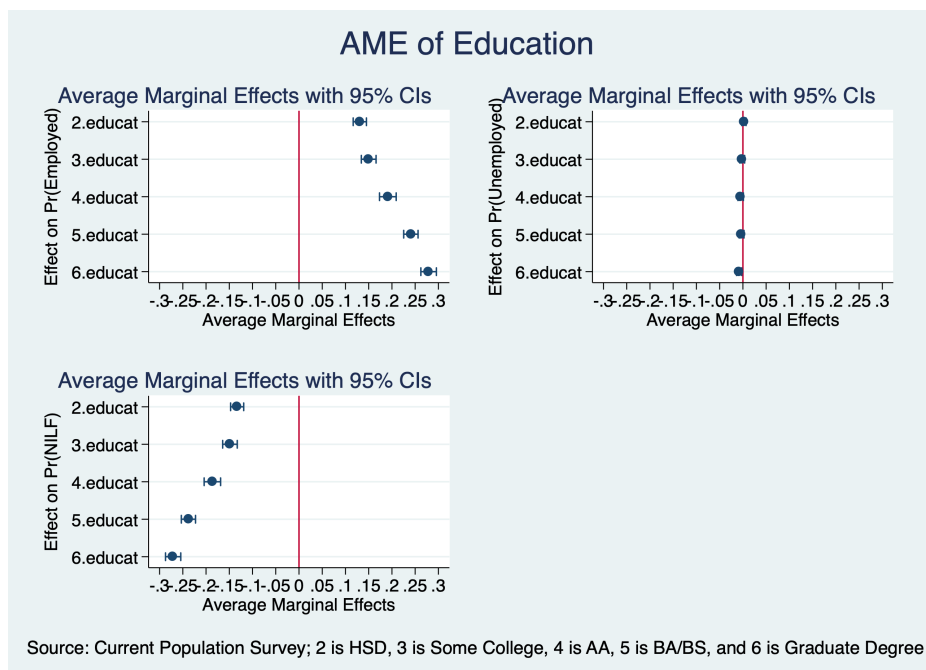


Figure 2.1: Marginal Effects of Education

```
graph drop Employed Unemployed NILF
```

Potential Experience

For Employed

```
margins, dydx(exp) at(exp=(0(2)60)) predict(outcome(1))
quietly marginsplot, allsimplelabels name(Employed) ytitle("Effect on Pr(Employed)") x
```

```
Average marginal effects          Number of obs      =      41,599
Model VCE      : OIM
```

```
Expression      : Pr(laborforcestatus==Employed), predict(outcome(1))
dy/dx w.r.t.    : exp
```

```
1._at          : exp              =              0
```

```
2._at          : exp              =              2
```

2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION43

3._at	: exp	=	4
4._at	: exp	=	6
5._at	: exp	=	8
6._at	: exp	=	10
7._at	: exp	=	12
8._at	: exp	=	14
9._at	: exp	=	16
10._at	: exp	=	18
11._at	: exp	=	20
12._at	: exp	=	22
13._at	: exp	=	24
14._at	: exp	=	26
15._at	: exp	=	28
16._at	: exp	=	30
17._at	: exp	=	32
18._at	: exp	=	34
19._at	: exp	=	36
20._at	: exp	=	38
21._at	: exp	=	40
22._at	: exp	=	42
23._at	: exp	=	44
24._at	: exp	=	46
25._at	: exp	=	48

26._at	:	exp	=	50
27._at	:	exp	=	52
28._at	:	exp	=	54
29._at	:	exp	=	56
30._at	:	exp	=	58
31._at	:	exp	=	60

		Delta-method					
		dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
exp	_at						
	1	.0119199	.0001127	105.73	0.000	.0116989	.0121408
	2	.0130781	.0001155	113.28	0.000	.0128519	.0133044
	3	.013976	.0001461	95.64	0.000	.0136896	.0142625
	4	.0145643	.0001796	81.10	0.000	.0142123	.0149163
	5	.0148298	.0002035	72.86	0.000	.0144308	.0152287
	6	.0147942	.000215	68.83	0.000	.0143729	.0152155
	7	.0145067	.0002152	67.40	0.000	.0140849	.0149285
	8	.0140316	.0002075	67.61	0.000	.0136249	.0144384
	9	.0134367	.0001953	68.81	0.000	.013054	.0138195
	10	.0127838	.0001811	70.60	0.000	.0124289	.0131387
	11	.0121231	.0001667	72.73	0.000	.0117965	.0124498
	12	.0114913	.000153	75.12	0.000	.0111915	.0117912
	13	.0109121	.0001403	77.76	0.000	.0106371	.0111872
	14	.0103981	.0001289	80.69	0.000	.0101455	.0106507
	15	.0099532	.0001186	83.95	0.000	.0097208	.0101855
	16	.0095751	.0001094	87.52	0.000	.0093606	.0097895
	17	.0092574	.0001014	91.30	0.000	.0090586	.0094561
	18	.0089913	.0000945	95.12	0.000	.008806	.0091766
	19	.0087672	.0000887	98.80	0.000	.0085933	.0089411
	20	.0085751	.0000839	102.22	0.000	.0084107	.0087395
	21	.0084059	.0000798	105.34	0.000	.0082495	.0085623
	22	.0082513	.0000762	108.24	0.000	.0081019	.0084007
	23	.0081038	.0000729	111.13	0.000	.0079609	.0082468
	24	.0079576	.0000697	114.23	0.000	.007821	.0080941
	25	.0078075	.0000663	117.83	0.000	.0076776	.0079373
	26	.0076498	.0000626	122.19	0.000	.0075271	.0077725
	27	.0074817	.0000586	127.62	0.000	.0073668	.0075967
	28	.0073016	.0000543	134.46	0.000	.0071952	.0074081

16._at	: exp	=	30
17._at	: exp	=	32
18._at	: exp	=	34
19._at	: exp	=	36
20._at	: exp	=	38
21._at	: exp	=	40
22._at	: exp	=	42
23._at	: exp	=	44
24._at	: exp	=	46
25._at	: exp	=	48
26._at	: exp	=	50
27._at	: exp	=	52
28._at	: exp	=	54
29._at	: exp	=	56
30._at	: exp	=	58
31._at	: exp	=	60

		Delta-method					
		dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
exp							
	_at						
	1	-.0125848	.0001155	-108.97	0.000	-.0128111	-.0123584
	2	-.0137412	.0001266	-108.55	0.000	-.0139893	-.013493
	3	-.0146118	.000159	-91.88	0.000	-.0149235	-.0143001
	4	-.0151489	.0001885	-80.35	0.000	-.0155185	-.0147794
	5	-.0153436	.0002048	-74.91	0.000	-.015745	-.0149421
	6	-.0152237	.0002061	-73.87	0.000	-.0156276	-.0148198
	7	-.0148454	.0001949	-76.17	0.000	-.0152274	-.0144634

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8		-.0142796	.0001756	-81.30	0.000	-.0146238	-.0139353
9		-.0135992	.0001528	-89.00	0.000	-.0138986	-.0132997
10		-.0128695	.0001299	-99.06	0.000	-.0131241	-.0126148
11		-.0121426	.0001093	-111.07	0.000	-.0123569	-.0119284
12		-.0114558	.0000922	-124.29	0.000	-.0116365	-.0112752
13		-.0108322	.0000787	-137.62	0.000	-.0109865	-.0106779
14		-.0102833	.0000686	-149.81	0.000	-.0104178	-.0101487
15		-.0098117	.0000614	-159.70	0.000	-.0099321	-.0096913
16		-.0094137	.0000566	-166.42	0.000	-.0095246	-.0093028
17		-.0090816	.0000536	-169.49	0.000	-.0091866	-.0089766
18		-.0088055	.0000521	-169.01	0.000	-.0089076	-.0087033
19		-.0085745	.0000517	-165.70	0.000	-.0086759	-.0084731
20		-.008378	.0000521	-160.73	0.000	-.0084802	-.0082759
21		-.0082063	.0000528	-155.31	0.000	-.0083098	-.0081027
22		-.0080503	.0000535	-150.38	0.000	-.0081552	-.0079453
23		-.0079024	.0000539	-146.55	0.000	-.0080081	-.0077967
24		-.0077563	.0000538	-144.13	0.000	-.0078617	-.0076508
25		-.0076067	.0000531	-143.27	0.000	-.0077108	-.0075027
26		-.0074499	.0000517	-144.07	0.000	-.0075512	-.0073485
27		-.0072829	.0000497	-146.60	0.000	-.0073802	-.0071855
28		-.007104	.000047	-151.07	0.000	-.0071962	-.0070118
29		-.0069124	.0000438	-157.78	0.000	-.0069983	-.0068266
30		-.0067085	.0000401	-167.14	0.000	-.0067872	-.0066298
31		-.0064935	.0000362	-179.60	0.000	-.0065643	-.0064226

Combine Graphs

```
graph combine Employed NILF, ycommon title("AME of Potential Experience")
graph export "/Users/Sam/Desktop/Econ 645/Stata/week8_mnlmexp.png", replace
```

```
graph drop Employed NILF
```

2.2.9 We can use coefplot with margins with eststo

```
quietly {est clear
eststo mnlm: quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.
}
```

Estimate the average marginal effects. Please note the post option when storing margin results

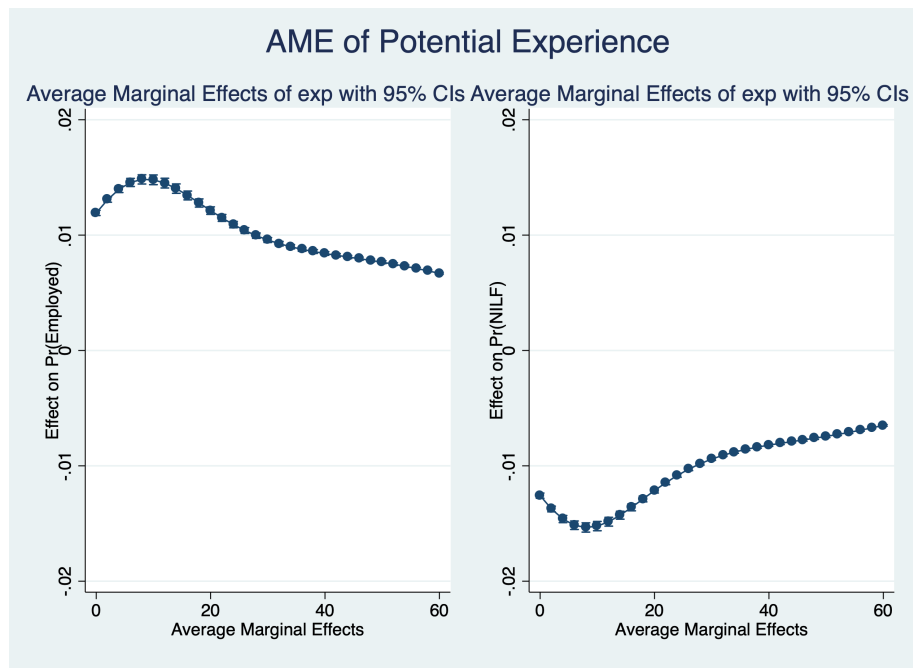


Figure 2.2: Marginal Effects of Potential Experience

```
quietly{
  eststo Employed: quietly margins, dydx(educat) predict(outcome(1)) post
  estimates restore mn1m
  eststo Unemployed: quietly margins, dydx(educat) predict(outcome(2)) post
  estimates restore mn2m
  eststo NILF: quietly margins, dydx(educat) predict(outcome(3)) post
}
```

Use coefplot

```
coefplot Employed Unemployed NILF, ///
  recast(bar) barw(0.15) vertical ///
  ciopts(recast(rcap) color(gs8)) citop ///
  xlab(1 "High School" 2 "Some College" 3 "Associates" 4 "Bachelor" 5 "Graduate") ///
  ytitle("Average Marginal Effect") ///
  xtitle("High Level of Education Attained") ///
  title("MNL: Probability of Labor Force Status", size(*0.7)) ///
  subtitle("By Education Relative to High School Dropout") ///
  caption("Source: Current Population Survey", size(*0.75)) ///
  name(coefplot3)
```


2.2. TEST FOR INDEPENDENCE OF IRRELEVANT ALTERNATIVE ASSUMPTION49

```
graph export "/Users/Sam/Desktop/Econ 645/Stata/week8_mnlmcoefplot.png", replace
```

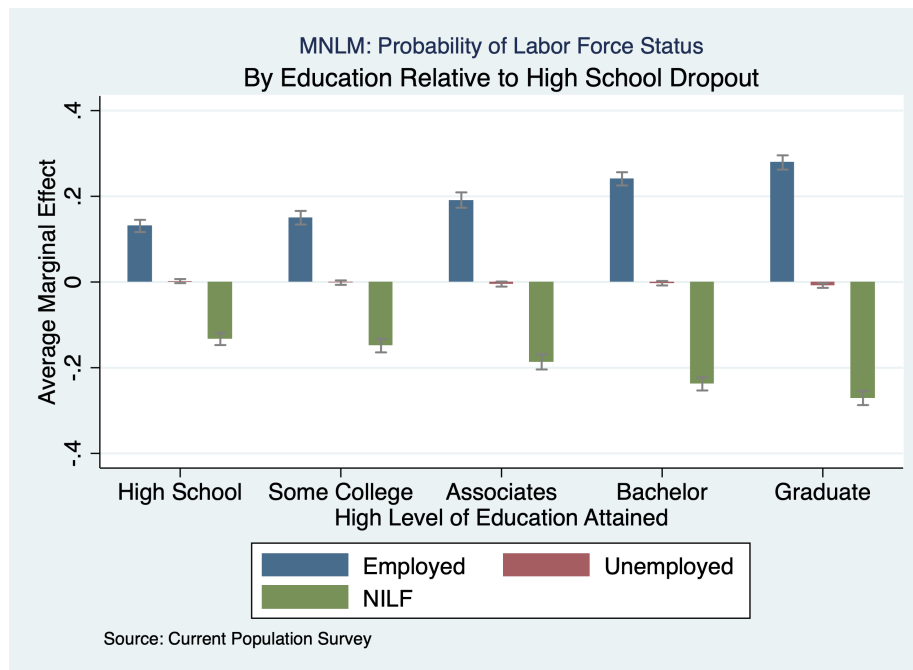


Figure 2.3: Using coefplot instead of marginsplot

```
graph drop coefplot3
```


Chapter 3

Ordinal Logit

Adapted from Long and Freese (2006)

3.1 Compare Ordinal Logit and Binary Logit

First, let's inspect the outcome of interest, which is labor force status.

```
use "/Users/Sam/Desktop/Econ 645/Data/binlfp2.dta", clear
tab lfp
```

(Data from 1976 PSID-T Mroz)

Paid Labor				
Force:				
1=yes 0=no		Freq.	Percent	Cum.
-----+				
NotInLF		325	43.16	43.16
inLF		428	56.84	100.00
-----+				
Total		753	100.00	

Now, let's compare our ordinal logit and binary logit coefficients when the outcome is binary.

```
quietly logit lfp k5 k618 age wc hc lwg inc, nolog
estimates store logit
quietly ologit lfp k5 k618 age wc hc lwg inc, nolog
```

```
estimates store ologit
estimates table logit ologit, b(%9.3f) t varlabel varwidth(30) equations(1:1)
```

	Variable	logit	ologit
-----+-----			
#1			
	# kids < 6	-1.463	-1.463
		-7.43	-7.43
	# kids 6-18	-0.065	-0.065
		-0.95	-0.95
	Wife's age in years	-0.063	-0.063
		-4.92	-4.92
	Wife College: 1=yes 0=no	0.807	0.807
		3.51	3.51
	Husband College: 1=yes 0=no	0.112	0.112
		0.54	0.54
	Log of wife's estimated wages	0.605	0.605
		4.01	4.01
	Family income excluding wife's	-0.034	-0.034
		-4.20	-4.20
	Constant	3.182	
		4.94	
-----+-----			
cut1			
	Constant		-3.182
			-4.94

legend: b/t

The coefficients are the same between the binary logit and the ordinal logit. The difference here is that the ordinal logit has a cut point and assumes an intercept of 0. The cutpoint and the intercept for binary logit are the same in absolute terms.

3.2 Example of Ordinal Dependent Variable

```
use "/Users/Sam/Desktop/Econ 645/Data/ordwarm2.dta", clear
describe warm yr89 male white age ed prst
```

(77 & 89 General Social Survey)

variable name	storage type	display format	value label	variable label
warm	byte	%10.0g	SD2SA	Mom can have warm relations with child
yr89	byte	%10.0g	yr1bl	Survey year: 1=1989 0=1977
male	byte	%10.0g	sex1bl	Gender: 1=male 0=female
white	byte	%10.0g	racelbl	Race: 1=white 0=not white
age	byte	%10.0g		Age in years
ed	byte	%10.0g		Years of education
prst	byte	%10.0g		Occupational prestige

```
summarize warm yr89 male white age ed prst
```

Variable	Obs	Mean	Std. Dev.	Min	Max
warm	2,293	2.607501	.9282156	1	4
yr89	2,293	.3986044	.4897178	0	1
male	2,293	.4648932	.4988748	0	1
white	2,293	.8765809	.3289894	0	1
age	2,293	44.93546	16.77903	18	89
ed	2,293	12.21805	3.160827	0	20
prst	2,293	39.58526	14.49226	12	82

Our outcome is the the survey results for people's opinion for a working mother

```
tab warm
```

Mom can			
have warm			
relations			
with child	Freq.	Percent	Cum.
SD	297	12.95	12.95
D	723	31.53	44.48
A	856	37.33	81.81
SA	417	18.19	100.00
Total	2,293	100.00	

We will run our ordinal logit by regressing opinion's on working mother's warm onto gender, race, age, survey year, education in years, and occupational prestige.

$$warm_{i,t} = \beta_1 male_i + \beta_2 race_i + \beta_3 age_{i,t} + \beta_4 edu_{i,t} + \beta_5 prestige_{i,t} + \beta_6 yr89_t + \varepsilon_{i,t}$$

```
ologit warm yr89 male white age ed prst, nolog
```

Ordered logistic regression	Number of obs	=	2,293
	LR chi2(6)	=	301.72
	Prob > chi2	=	0.0000
Log likelihood = -2844.9123	Pseudo R2	=	0.0504

	warm	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
yr89		.5239025	.0798989	6.56	0.000	.3673036	.6805014
male		-.7332997	.0784827	-9.34	0.000	-.887123	-.5794765
white		-.3911595	.1183808	-3.30	0.001	-.6231816	-.1591373
age		-.0216655	.0024683	-8.78	0.000	-.0265032	-.0168278
ed		.0671728	.015975	4.20	0.000	.0358624	.0984831
prst		.0060727	.0032929	1.84	0.065	-.0003813	.0125267
/cut1		-2.465362	.2389128			-2.933622	-1.997102
/cut2		-.630904	.2333156			-1.088194	-.1736138
/cut3		1.261854	.234018			.8031871	1.720521

We have three cutpoints

$$\tau_1 = -2.47, \tau_2 = -0.63, \tau_3 = 1.26$$

3.3 Test Proportional Odds Assumption

We have our two tests for testing the proportional odds assumption 1. Approximate Likelihood Ratio Test 2. Wald Test

3.3.1 Approximate Likelihood Ratio Test

We use the *omodel* command to run an Approximate Likelihood Ratio Test

```
omodel logit warm yr89 male white age ed prst
```

```

Iteration 0:  log likelihood = -2995.7704
Iteration 1:  log likelihood = -2846.4532
Iteration 2:  log likelihood = -2844.9142
Iteration 3:  log likelihood = -2844.9123

```

```

Ordered logit estimates                                Number of obs   =      2293
                                                       LR chi2(6)      =      301.72
                                                       Prob > chi2     =      0.0000
Log likelihood = -2844.9123                          Pseudo R2      =      0.0504

```

	warm	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
yr89		.5239025	.0798988	6.56	0.000	.3673037	.6805013
male		-.7332997	.0784827	-9.34	0.000	-.8871229	-.5794766
white		-.3911595	.1183808	-3.30	0.001	-.6231815	-.1591374
age		-.0216655	.0024683	-8.78	0.000	-.0265032	-.0168278
ed		.0671728	.015975	4.20	0.000	.0358624	.0984831
prst		.0060727	.0032929	1.84	0.065	-.0003813	.0125267
-----+-----							
_cut1		-2.465362	.2389126	(Ancillary parameters)			
_cut2		-.630904	.2333155				
_cut3		1.261854	.2340179				

Approximate likelihood-ratio test of proportionality of odds
across response categories:

```

      chi2(12) =      48.91
      Prob > chi2 =      0.0000

```

With the `oparallel` package

```

help oparallel
quietly omodel logit warm yr89 male white age ed prst
oparallel, lr

```

Test of the parallel regression assumption

	Chi2	df	P>Chi2
likelihood ratio	49.2	12	0.000

We reject the null hypothesis that the coefficients are the same across $1, \dots, J-1$ equations at the 0.1% level.

3.3.2 Wald Test

We use the *brant* command for the Wald Test

```
quietly ologit warm yr89 male white age ed prst, nolog
oparallel, lr
```

```
quietly use "/Users/Sam/Desktop/Econ 645/Data/ordwarm2.dta", clear
quietly ologit warm yr89 male white age ed prst, nolog
oparallel, lr
```

We reject the null hypothesis that the coefficients are the same across $1, \dots, J-1$ equations at the 0.1% level.

3.4 Interpretation

3.4.1 Odds Ratios

```
ologit warm yr89 male white age ed prst, or
oparallel, lr
listcoef, help
listcoef, help percent
```

```
Iteration 0:  log likelihood = -2995.7704
Iteration 1:  log likelihood = -2846.4532
Iteration 2:  log likelihood = -2844.9142
Iteration 3:  log likelihood = -2844.9123
Iteration 4:  log likelihood = -2844.9123
```

Ordered logistic regression	Number of obs	=	2,293
	LR chi2(6)	=	301.72
	Prob > chi2	=	0.0000
Log likelihood = -2844.9123	Pseudo R2	=	0.0504

	warm	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]
yr89		1.688605	.1349176	6.56	0.000	1.443836 1.974868
male		.4803214	.0376969	-9.34	0.000	.4118389 .5601916
white		.6762723	.0800577	-3.30	0.001	.5362356 .8528792
age		.9785675	.0024154	-8.78	0.000	.9738449 .983313
ed		1.06948	.0170849	4.20	0.000	1.036513 1.103496

prst	1.006091	.003313	1.84	0.065	.9996188	1.012605
-----+						
/cut1	-2.465362	.2389128			-2.933622	-1.997102
/cut2	-.630904	.2333156			-1.088194	-.1736138
/cut3	1.261854	.234018			.8031871	1.720521

ologit (N=2293): Factor Change in Odds

Odds of: >m vs <=m

warm	b	z	P> z	e ^b	e ^b StdX	SDofX
-----+						
yr89	0.52390	6.557	0.000	1.6886	1.2925	0.4897
male	-0.73330	-9.343	0.000	0.4803	0.6936	0.4989
white	-0.39116	-3.304	0.001	0.6763	0.8792	0.3290
age	-0.02167	-8.778	0.000	0.9786	0.6952	16.7790
ed	0.06717	4.205	0.000	1.0695	1.2365	3.1608
prst	0.00607	1.844	0.065	1.0061	1.0920	14.4923

b = raw coefficient

z = z-score for test of b=0

P>|z| = p-value for z-test

e^b = exp(b) = factor change in odds for unit increase in X

e^bStdX = exp(b*SD of X) = change in odds for SD increase in X

SDofX = standard deviation of X

ologit (N=2293): Percentage Change in Odds

Odds of: >m vs <=m

warm	b	z	P> z	%	%StdX	SDofX
-----+						
yr89	0.52390	6.557	0.000	68.9	29.2	0.4897
male	-0.73330	-9.343	0.000	-52.0	-30.6	0.4989
white	-0.39116	-3.304	0.001	-32.4	-12.1	0.3290
age	-0.02167	-8.778	0.000	-2.1	-30.5	16.7790
ed	0.06717	4.205	0.000	6.9	23.7	3.1608
prst	0.00607	1.844	0.065	0.6	9.2	14.4923

b = raw coefficient

z = z-score for test of b=0

For a male, the odds of a **negative attitude** toward working mothers increase by a factor of $e^{(-0.73330)} = 2.08$, *ceteris paribus*

3.4.2 Average Marginal Effects

Ordered logistic regression	Number of obs	=	2,293
	LR chi2(6)	=	301.72
	Prob > chi2	=	0.0000
Log likelihood = -2844.9123	Pseudo R2	=	0.0504

	warm	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]
	yr89	1.688605	.1349176	6.56	0.000	1.443836 1.974868
	male	.4803214	.0376969	-9.34	0.000	.4118389 .5601916
	white	.6762723	.0800577	-3.30	0.001	.5362356 .8528792
	age	.9785675	.0024154	-8.78	0.000	.9738449 .983313
	ed	1.06948	.0170849	4.20	0.000	1.036513 1.103496
	prst	1.006091	.003313	1.84	0.065	.9996188 1.012605

3.4. INTERPRETATION

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/cut1	-2.465362	.2389128	-2.933622	-1.997102
/cut2	-.630904	.2333156	-1.088194	-.1736138
/cut3	1.261854	.234018	.8031871	1.720521

Average marginal effects Number of obs = 2,293
Model VCE : OIM

```
Expression      : Pr(warm==1), predict(outcome(1))
dy/dx w.r.t.   : yr89 male white age ed prst
```

	Delta-method					
	dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
yr89	-.0557094	.0087921	-6.34	0.000	-.0729416	-.0384771
male	.0779757	.0088072	8.85	0.000	.0607139	.0952375
white	.0415941	.0127155	3.27	0.001	.0166721	.0665161
age	.0023038	.0002754	8.37	0.000	.0017641	.0028435
ed	-.0071428	.0017147	-4.17	0.000	-.0105036	-.0037821
prst	-.0006457	.0003514	-1.84	0.066	-.0013345	.000043

Average marginal effects Number of obs = 2,293
Model VCE : OIM

```
Expression      : Pr(warm==2), predict(outcome(2))
dy/dx w.r.t.   : yr89 male white age ed prst
```

	Delta-method					
	dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
yr89	-.0608201	.0091774	-6.63	0.000	-.0788075	-.0428328
male	.0851292	.0089196	9.54	0.000	.0676472	.1026112
white	.0454099	.0136969	3.32	0.001	.0185644	.0722554
age	.0025152	.0002817	8.93	0.000	.001963	.0030673
ed	-.0077981	.00185	-4.22	0.000	-.0114241	-.0041722
prst	-.000705	.0003818	-1.85	0.065	-.0014533	.0000433

Average marginal effects	Number of obs	=	2,293
Model VCE : OIM			

Expression : Pr(warm==3), predict(outcome(3))
 dy/dx w.r.t. : yr89 male white age ed prst

		Delta-method				[95% Conf. Interval]	
		dy/dx	Std. Err.	z	P> z		
yr89		.043531	.0069122	6.30	0.000	.0299833	.0570787
male		-.0609298	.0067839	-8.98	0.000	-.074226	-.0476336
white		-.0325014	.0099733	-3.26	0.001	-.0520487	-.0129541
age		-.0018002	.0002138	-8.42	0.000	-.0022193	-.0013811
ed		.0055814	.0013342	4.18	0.000	.0029665	.0081963
prst		.0005046	.0002752	1.83	0.067	-.0000349	.0010441

Average marginal effects Number of obs = 2,293
 Model VCE : OIM

Expression : Pr(warm==4), predict(outcome(4))
 dy/dx w.r.t. : yr89 male white age ed prst

		Delta-method				[95% Conf. Interval]	
		dy/dx	Std. Err.	z	P> z		
yr89		.0729985	.0111797	6.53	0.000	.0510866	.0949103
male		-.102175	.0111936	-9.13	0.000	-.1241141	-.080236
white		-.0545026	.016485	-3.31	0.001	-.0868126	-.0221926
age		-.0030188	.0003501	-8.62	0.000	-.0037051	-.0023325
ed		.0093596	.0022406	4.18	0.000	.0049681	.0137511
prst		.0008461	.0004584	1.85	0.065	-.0000522	.0017445

3.4.3 Marginal Effects at the Average

```
ologit warm yr89 male white age ed prst, or
margins, dydx(*) atmeans predict(outcome(1))
margins, dydx(*) atmeans predict(outcome(2))
margins, dydx(*) atmeans predict(outcome(3))
margins, dydx(*) atmeans predict(outcome(4))
```

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Iteration 0: log likelihood = -2995.7704
 Iteration 1: log likelihood = -2846.4532
 Iteration 2: log likelihood = -2844.9142
 Iteration 3: log likelihood = -2844.9123
 Iteration 4: log likelihood = -2844.9123

Ordered logistic regression	Number of obs	=	2,293
	LR chi2(6)	=	301.72
	Prob > chi2	=	0.0000
Log likelihood = -2844.9123	Pseudo R2	=	0.0504

	warm	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
yr89		1.688605	.1349176	6.56	0.000	1.443836	1.974868
male		.4803214	.0376969	-9.34	0.000	.4118389	.5601916
white		.6762723	.0800577	-3.30	0.001	.5362356	.8528792
age		.9785675	.0024154	-8.78	0.000	.9738449	.983313
ed		1.06948	.0170849	4.20	0.000	1.036513	1.103496
prst		1.006091	.003313	1.84	0.065	.9996188	1.012605
/cut1		-2.465362	.2389128			-2.933622	-1.997102
/cut2		-.630904	.2333156			-1.088194	-.1736138
/cut3		1.261854	.234018			.8031871	1.720521

Conditional marginal effects	Number of obs	=	2,293
Model VCE : OIM			

Expression : Pr(warm==1), predict(outcome(1))
 dy/dx w.r.t. : yr89 male white age ed prst
 at : yr89 = .3986044 (mean)
 male = .4648932 (mean)
 white = .8765809 (mean)
 age = 44.93546 (mean)
 ed = 12.21805 (mean)
 prst = 39.58526 (mean)

		Delta-method				[95% Conf. Interval]	
		dy/dx	Std. Err.	z	P> z		
yr89		-.051803	.0081383	-6.37	0.000	-.0677537	-.0358522
male		.0725079	.0081294	8.92	0.000	.0565746	.0884413
white		.0386775	.0118016	3.28	0.001	.0155468	.0618081

Delta-method				
dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]

	yr89	.0582103	.0096161	6.05	0.000	.0393631	.0770575
	male	-.0814762	.0099979	-8.15	0.000	-.1010717	-.0618807
	white	-.0434613	.0134313	-3.24	0.001	-.0697862	-.0171364
	age	-.0024072	.0003126	-7.70	0.000	-.0030199	-.0017945
	ed	.0074635	.0018362	4.06	0.000	.0038645	.0110625
	prst	.0006747	.0003683	1.83	0.067	-.0000472	.0013967

Conditional marginal effects	Number of obs	=	2,293
Model VCE : OIM			

```

Expression      : Pr(warm==4), predict(outcome(4))
dy/dx w.r.t.   : yr89 male white age ed prst
at
  yr89          =      .3986044 (mean)
  male          =      .4648932 (mean)
  white         =      .8765809 (mean)
  age           =     44.93546 (mean)
  ed            =     12.21805 (mean)
  prst         =     39.58526 (mean)

```

	Delta-method					
	dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
yr89	.0708428	.0109038	6.50	0.000	.0494718	.0922139
male	-.0991578	.0109073	-9.09	0.000	-.1205356	-.0777799
white	-.0528931	.0160474	-3.30	0.001	-.0843454	-.0214409
age	-.0029296	.00034	-8.62	0.000	-.0035961	-.0022632
ed	.0090832	.0021702	4.19	0.000	.0048296	.0133368
prst	.0008212	.0004455	1.84	0.065	-.0000519	.0016943

Chapter 4

Reshape

From my experience, I feel that understanding and mastering the reshape command is invaluable. This is especially true when working with panel data. Panel data should not be used in a wide format. Panel data should be in a long format where you can use the xtset unit time command. After the xtset is established you can easily add lags or differences with the l.var or d.var

It is also helpful in terms of merging datasets. A lot of time if I'm merging BLS data, such as state unemployment, or inflation rates, I need to reshape the BLS data from wide to long to I can merge m:1

I typically work in long formats, but reshaping to wide formats does have its benefits. I find wide formats are helpful for people who work in excel, especially when years are across columns.

4.1 Wide and Long Datasets

We have two types of datasets: wide and long

Wide Format

Our wide datasets typically include years or variables types spread out across the columns.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"  
use cardio_wide, clear  
describe
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

Contains data from `cardio_wide.dta`

```
obs:      6
vars:     12
size:     120
```

22 Dec 2009 20:43

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
age	byte	%3.0f		Age of person
bp1	int	%3.0f		Blood pressure systolic Trial 1
bp2	int	%3.0f		Blood pressure systolic Trial 2
bp3	int	%3.0f		Blood pressure systolic Trial 3
bp4	int	%3.0f		Blood pressure systolic Trial 4
bp5	int	%3.0f		Blood pressure systolic Trial 5
p11	int	%3.0f		Pulse: Trial 1
p12	byte	%3.0f		Pulse: Trial 2
p13	int	%3.0f		Pulse: Trial 3
p14	int	%3.0f		Pulse: Trial 4
p15	byte	%3.0f		Pulse: Trial 5

Sorted by:

We have 12 variables `idcode`, `age`, and our 5 different blood pressure and pulse trials. We see that are 5 different incidents/trials of blood pressure and pulse are spread out across 5 different variables `bp1 – bp5` and `p11 – p15`*

`list`

	id	age	bp1	bp2	bp3	bp4	bp5	p11	p12	p13	p14	p15
1.	1	40	115	86	129	105	127	54	87	93	81	92
2.	2	30	123	136	107	111	120	92	88	125	87	58
3.	3	16	124	122	101	109	112	105	97	128	57	68
4.	4	23	105	115	121	129	137	52	79	71	106	39
5.	5	18	116	128	112	125	111	70	64	52	68	59
6.	6	27	108	126	124	131	107	74	78	92	99	80

With a wide format each row is an observation and all of our different trials are located on a single row

Long Format

We'll look at an example of the same dataset but in long format

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
describe
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

Contains data from cardio_long.dta

```
obs:      30
vars:      5                      10 Feb 2020 23:02
size:     210
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
trial	byte	%9.0g		Trial number
age	byte	%3.0f		Age of person
bp	int	%3.0f		Blood pressure (systolic)
pl	int	%3.0f		Pulse

Sorted by: id trial

We only have 5 variables idcode, age, trial, blood pressure, and pulse.

```
list
```

	id	trial	age	bp	pl
1.	1	1	40	115	54
2.	1	2	40	86	87
3.	1	3	40	129	93
4.	1	4	40	105	81
5.	1	5	40	127	92
6.	2	1	30	123	92
7.	2	2	30	136	88
8.	2	3	30	107	125
9.	2	4	30	111	87
10.	2	5	30	120	58

11.		3		1	
12.		3		2	
13.		3		3	
14.		3		4	
15.		3		5	

16.		4		1	
17.		4		2	
18.		4		3	
19.		4		4	
20.		4		5	

21.		5		1	
22.		5		2	
23.		5		3	
24.		5		4	
25.		5		5	

26.		6		1	
27.		6		2	
28.		6		3	
29.		6		4	
30.		6		5	
+-----+					

We also see that each individual has 5 observations. One for each trial that is sorted from 1 to 5.

Notice: We have a cross-sectional unit variable with *idcode* and we also have a time component with *trial*. Our dataset is basically a panel data set. We observe the same person over 5 trials.

What kind of format should we use?

It depends.

Wide Format

Mitchell makes a good point of looking at correlations among the blood pressure trials.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide, clear
correlate bp1 bp2 bp3 bp4 bp5
```

/Users/Sam/Desktop/Econ 645/Data/Mitchell

(obs=6)

		bp1	bp2	bp3	bp4	bp5
bp1		1.0000				
bp2		0.2427	1.0000			
bp3		-0.7662	-0.6657	1.0000		
bp4		-0.7644	0.3980	0.2644	1.0000	
bp5		-0.3643	-0.4984	0.3694	-0.0966	1.0000

Mitchell also mentions that multivariate analysis with multiple trials is a bit easier with factor variables. If we wanted to see all of the different regressions for each trial, then

```
mvreg bp* = age
```

Equation		Obs	Parms	RMSE	"R-sq"	F	P
bp1		6	2	8.516293	0.0160	.0651388	0.8111
bp2		6	2	15.34222	0.3858	2.512064	0.1882
bp3		6	2	8.846404	0.4597	3.402782	0.1388
bp4		6	2	11.60081	0.1554	.7357661	0.4394
bp5		6	2	11.81248	0.1360	.629676	0.4719

		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
bp1						
	age	-.1107266	.4338427	-0.26	0.811	-1.315267 1.093814
	_cons	118.0087	11.66545	10.12	0.001	85.62017 150.3971
bp2						
	age	-1.238754	.7815736	-1.58	0.188	-3.40875 .9312418
	_cons	150.628	21.01547	7.17	0.002	92.27974 208.9763
bp3						
	age	.8313149	.4506595	1.84	0.139	-.4199164 2.082546
	_cons	94.32958	12.11763	7.78	0.001	60.68565 127.9735
bp4						
	age	-.5069204	.5909761	-0.86	0.439	-2.147733 1.133892
	_cons	131.3443	15.89055	8.27	0.001	87.22504 175.4635

-----+-----							
bp5							
age		.4775087	.6017591	0.79	0.472	-1.193242	2.14826
_cons		106.7439	16.1805	6.60	0.003	61.81969	151.6682
-----+-----							

Long Format

There are several reasons we would want to use long format over wide format

Correlations

If we want correlations between blood pressure and pulse, a long format is preferable.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
correlate bp pl
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
(obs=30)
```

		bp	pl
-----+-----			
bp		1.0000	
pl		-0.0444	1.0000

Or

```
bysort trial: correlate bp pl
```

```
-> trial = 1
```

```
(obs=6)
```

		bp	pl
-----+-----			
bp		1.0000	
pl		0.7958	1.0000

```
-> trial = 2
(obs=6)
```

		bp	pl
bp		1.0000	
pl		-0.1985	1.0000

```
-> trial = 3
(obs=6)
```

		bp	pl
bp		1.0000	
pl		-0.4790	1.0000

```
-> trial = 4
(obs=6)
```

		bp	pl
bp		1.0000	
pl		0.5639	1.0000

```
-> trial = 5
(obs=6)
```

		bp	pl
bp		1.0000	
pl		-0.3911	1.0000

Panel data

It is also more appropriate for panel data and the `xt` commands, such as `xtset`, `xtreg`, etc. I also prefer this for regular regressions where we can use factor variables `i.var` for different time periods.

```
xtset id trial
xtreg bp age
xtline bp, overlay
```

Recoding

It is also easier to recode in long format.

```
recode pl (min/89=0) (90/max=1), generate (plhi)
```

Recoding among many variables in wide format is more of a pain. You can see this on page 293 in Mitchell.

Egen is easier

If you wanted to mean of each 5 trials, it just as easy with egen in long format.

```
bysort id: egen pl_avg=mean(pl)
bysort id: egen bp_avg=mean(bp)
```

For wide-format you can use rowmean with the wildcard

```
use cardio_wide, clear
```

We don't need the bysort group, but we need to be careful which columns to include

```
egen pl_avg = rowmean(pl*)
egen bp_avg = rowmean(bp*)
```

It is easier to explicitly difference trials with wide-format, but requires more typing.

```
generate pldiff2 = p12-p11
generate pldiff3 = p13-p12
generate pldiff4 = p14-p13
generate pldiff5 = p15-p14
```


4.1.0.1 Using subscripting/indexing, differences, lags with ease

I prefer using subscripting/indexing within groups or using xtset with panel data since we have a lot of flexibilities.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
sort id trial
```

We can use indexing

```
by id: generate pldiff = p1-p1[_n-1]
```

We can difference from the first trial

```
by id: generate pldiff_1trial = p1-p1[1]
```

We can use xtset as well with l.var

```
xtset id trial
```

```
      panel variable:  id (strongly balanced)
      time variable:  trial, 1 to 5
                   delta:  1 unit
```

One lag

```
gen pldiff_other = p1-l.p1
```

We can use two lags

```
gen pldiff2_other = l2.p1-l1.p1
```

We can use three lags

```
gen pldiff3_other = l3.p1-l2.p1
list
```

(18 missing values generated)

```
+-----+
| id   trial   age   bp    pl    pldiff   pldiff~1   ~f_other   p~2_ot~r   p~3_ot~r |
```

1.		1	1	40	115	54	.	0	.	.	
2.		1	2	40	86	87	33	33	33	.	
3.		1	3	40	129	93	6	39	6	-33	
4.		1	4	40	105	81	-12	27	-12	-6	-3
5.		1	5	40	127	92	11	38	11	12	
6.		2	1	30	123	92	.	0	.	.	
7.		2	2	30	136	88	-4	-4	-4	.	
8.		2	3	30	107	125	37	33	37	4	
9.		2	4	30	111	87	-38	-5	-38	-37	
10.		2	5	30	120	58	-29	-34	-29	38	-3
11.		3	1	16	124	105	.	0	.	.	
12.		3	2	16	122	97	-8	-8	-8	.	
13.		3	3	16	101	128	31	23	31	8	
14.		3	4	16	109	57	-71	-48	-71	-31	
15.		3	5	16	112	68	11	-37	11	71	-3
16.		4	1	23	105	52	.	0	.	.	
17.		4	2	23	115	79	27	27	27	.	
18.		4	3	23	121	71	-8	19	-8	-27	
19.		4	4	23	129	106	35	54	35	8	-2
20.		4	5	23	137	39	-67	-13	-67	-35	
21.		5	1	18	116	70	.	0	.	.	
22.		5	2	18	128	64	-6	-6	-6	.	
23.		5	3	18	112	52	-12	-18	-12	6	
24.		5	4	18	125	68	16	-2	16	12	
25.		5	5	18	111	59	-9	-11	-9	-16	:
26.		6	1	27	108	74	.	0	.	.	
27.		6	2	27	126	78	4	4	4	.	
28.		6	3	27	124	92	14	18	14	-4	
29.		6	4	27	131	99	7	25	7	-14	-
30.		6	5	27	107	80	-19	6	-19	-7	-5

Wide format can become unwieldy quickly and Stata does have a variable limit size, while your observation limit is typically your memory. In short, it is better in the long-run to learn to work with Stata in long-format.

However, Mitchell does bring up a good point that in our panel data, we may need multiple weights `wt1-wtk`. You can see this in the ACS PUMS with 250 or so replicate weights. We do not want to reshape our data by weights, we need to keep the data in a cross-sectional unit `i` and time period `t` format.

In short, I agree with Mitchell that you should learn to work with your data in a long-format data structure.

4.2 Reshaping Long to Wide

There may be scenarios where we need to reshape our long-format data into a wide format. For example, it might be easier to match and merge our data if our using data set is in wide format and our master data set is in wide format

Let's look at an example

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
describe
list
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

Contains data from cardio_long.dta

```
obs:      30
vars:      5                               10 Feb 2020 23:02
size:      210
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
trial	byte	%9.0g		Trial number
age	byte	%3.0f		Age of person
bp	int	%3.0f		Blood pressure (systolic)
pl	int	%3.0f		Pulse

Sorted by: id trial

	id	trial	age	bp	pl
1.	1	1	40	115	54
2.	1	2	40	86	87
3.	1	3	40	129	93
4.	1	4	40	105	81
5.	1	5	40	127	92

Our reshape command requires *reshape* 1. What direction: wide or long 2. Variables to reshape (bp and pl) 3. Variables define an observation i(id) 4. Variables that defines the repeated observations for each person orvtime variable j(trial) for reshaping long to wide

```
reshape wide bp pl, i(id) j(trial)
describe
list
```

Data long -> wide

```

-----
Number of obs.                30  ->      6
Number of variables            5  ->     12
j variable (5 values)         trial -> (dropped)
xij variables:
                                bp  ->  bp1 bp2 ... bp5
                                pl  ->  pl1 pl2 ... pl5
-----

```

Contains data

```

obs:      6
vars:     12
size:    132

```

```

-----
      storage  display  value
variable name  type    format  label      variable label
-----
id             byte    %3.0f
bp1            int     %3.0f      1 bp
pl1            int     %3.0f      1 pl
bp2            int     %3.0f      2 bp
pl2            int     %3.0f      2 pl
bp3            int     %3.0f      3 bp
pl3            int     %3.0f      3 pl
bp4            int     %3.0f      4 bp
pl4            int     %3.0f      4 pl
bp5            int     %3.0f      5 bp
pl5            int     %3.0f      5 pl
age            byte    %3.0f
Age of person
-----

```

Sorted by: id

Note: Dataset has changed since last saved.

```

+-----+
| id  bp1  pl1  bp2  pl2  bp3  pl3  bp4  pl4  bp5  pl5  age |
+-----+
1. | 1  115   54   86   87  129   93  105   81  127   92   40 |
2. | 2  123   92  136   88  107  125  111   87  120   58   30 |
3. | 3  124  105  122   97  101  128  109   57  112   68   16 |
4. | 4  105   52  115   79  121   71  129  106  137   39   23 |
5. | 5  116   70  128   64  112   52  125   68  111   59   18 |
+-----+
6. | 6  108   74  126   78  124   92  131   99  107   80   27 |
+-----+

```

Notice we excluded age. It is constant within the cross-sectional unit. It does not define the cross-sectional unit or the*time dimension unit. We do not want to separate it into repeated observations either.

Without doing any additional work, we can reshape it back to long with simply reshape long.

```
reshape long
describe
```

(note: j = 1 2 3 4 5)

Data	wide	->	long
Number of obs.	6	->	30
Number of variables	12	->	5
j variable (5 values)		->	trial
xij variables:			
	bp1 bp2 ... bp5	->	bp
	pl1 pl2 ... pl5	->	pl

Contains data

```
obs:      30
vars:      5
size:     210
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
trial	byte	%9.0g		Trial number
bp	int	%3.0f		Blood pressure (systolic)
pl	int	%3.0f		Pulse
age	byte	%3.0f		Age of person

Sorted by: id trial

Note: Dataset has changed since last saved.

4.3 Reshaping Long to Wide: Problems

Problem 1: A constant value within a cross-sectional unit is not constant

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long2, clear
list in 1/10
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

	id	trial	age	bp	pl
1.	1	1	40	115	54
2.	1	2	40	86	87
3.	1	3	44	129	93
4.	1	4	40	105	81
5.	1	5	40	127	92
6.	2	1	30	123	92
7.	2	2	30	136	88
8.	2	3	30	107	125
9.	2	4	30	111	87
10.	2	5	30	120	58

We see that cross-sectional unit 1 has a data entry problem with age in the 3rd trial. If we reshape, we will get an error

```
reshape wide bp pl, i(id) j(trial)
```

(note: j = 1 2 3 4 5)

variable age not constant within id

Your data are currently long. You are performing a reshape wide. You typed something like

```
. reshape wide a b, i(id) j(trial)
```

There are variables other than a, b, id, trial in your data. They must be constant within id. The only way they can fit into wide data without loss of information.

The variable or variables listed above are not constant within id. Perhaps the values are incorrect. Use the `list` command to get a list of the problem observations.

Either that, or the values vary because they should vary, in which case you must exclude the list of xij variables to be reshaped, or drop them.

```
r(9);

r(9);
```

Use the reshape error command to help with the problem

```
reshape error
```

It tells us that our constant variable is not constant within cross-sectional units.

Problem 2: Excluding a non-constant that we need to reshape

If we excluded pl from our reshape, we will get a similar error.

```
capture reshape wide bp, i(id) j(trial)
reshape error
```

(note: j = 1 2 3 4 5)

i (id) indicates the top-level grouping such as subject id.

j (trial) indicates the subgrouping such as time.

xij variable is bp.

Thus, the following variable(s) should be constant within i:

```
age pl
```

age not constant within i (id) for 1 value of i:

```
+-----+
| id   trial   age |
+-----+
1. | 1       1   40 |
2. | 1       2   40 |
3. | 1       3   44 |
4. | 1       4   40 |
5. | 1       5   40 |
+-----+
```

pl not constant within i (id) for 6 values of i:

	id	trial	pl
1.	1	1	54
2.	1	2	87
3.	1	3	93
4.	1	4	81
5.	1	5	92
6.	2	1	92
7.	2	2	88
8.	2	3	125
9.	2	4	87
10.	2	5	58
11.	3	1	105
12.	3	2	97
13.	3	3	128
14.	3	4	57
15.	3	5	68
16.	4	1	52
17.	4	2	79
18.	4	3	71
19.	4	4	106
20.	4	5	39
21.	5	1	70
22.	5	2	64
23.	5	3	52
24.	5	4	68
25.	5	5	59
26.	6	1	74
27.	6	2	78
28.	6	3	92
29.	6	4	99
30.	6	5	80

(data now sorted by id trial)

4.4 Reshaping Wide to Long

In my personal opinion, this reshaping wide to long is a more common occurrence. It is important to get data into a long-format so we can conduct panel analysis. I had to do this for a few of the Wooldridge datasets that were panel data, but in wide format.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide, clear
describe
list
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

Contains data from cardio_wide.dta

```
obs:      6
vars:     12
size:     120
22 Dec 2009 20:43
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
age	byte	%3.0f		Age of person
bp1	int	%3.0f		Blood pressure systolic Trial 1
bp2	int	%3.0f		Blood pressure systolic Trial 2
bp3	int	%3.0f		Blood pressure systolic Trial 3
bp4	int	%3.0f		Blood pressure systolic Trial 4
bp5	int	%3.0f		Blood pressure systolic Trial 5
p11	int	%3.0f		Pulse: Trial 1
p12	byte	%3.0f		Pulse: Trial 2
p13	int	%3.0f		Pulse: Trial 3
p14	int	%3.0f		Pulse: Trial 4
p15	byte	%3.0f		Pulse: Trial 5

Sorted by:

	id	age	bp1	bp2	bp3	bp4	bp5	p11	p12	p13	p14	p15
1.	1	40	115	86	129	105	127	54	87	93	81	92
2.	2	30	123	136	107	111	120	92	88	125	87	58
3.	3	16	124	122	101	109	112	105	97	128	57	68

4.		4	23	105	115	121	129	137	52	79	71	106	39	
5.		5	18	116	128	112	125	111	70	64	52	68	59	

6.		6	27	108	126	124	131	107	74	78	92	99	80	

Let's reshape wide to long.

Our reshape values that are not constant: bp and pl.

Our cross-sectional unit to reshape upon or our list of variables that uniquely identify a cross-sectional unit. For example, we may need a State FIPS code and a County FIPS code to unique identify a county cross-sectional unit (of course we can merge the 2-digit State FIPS code and the 3-digit County FIPS code to create a 5-digit FIPS code in one variable).

Our time unit, where with reshape long, we need to create a new variable that takes on the values of the suffixes of bp and pl (For example trialnum). Our constant values within cross-sectional units do not need to be identified.

```
reshape long bp pl, i(id) j(trialnum)
describe
list in 1/10
```

(note: j = 1 2 3 4 5)

Data	wide	->	long
Number of obs.	6	->	30
Number of variables	12	->	5
j variable (5 values)		->	trialnum
xij variables:			
	bp1 bp2 ... bp5	->	bp
	pl1 pl2 ... pl5	->	pl

Contains data

```
obs:      30
vars:      5
size:     210
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
trialnum	byte	%9.0g		

```

age          byte    %3.0f          Age of person
bp           int     %3.0f
pl           int     %3.0f

```

Sorted by: id trialnum

Note: Dataset has changed since last saved.

```

+-----+
| id  trialnum  age  bp  pl |
+-----+
1. | 1      1    40  115  54 |
2. | 1      2    40   86  87 |
3. | 1      3    40  129  93 |
4. | 1      4    40  105  81 |
5. | 1      5    40  127  92 |
+-----+
6. | 2      1    30  123  92 |
7. | 2      2    30  136  88 |
8. | 2      3    30  107 125 |
9. | 2      4    30  111  87 |
10. | 2      5    30  120  58 |
+-----+

```

To reshape it back to wide, we just simply state

```

reshape wide
describe
list

```

(note: j = 1 2 3 4 5)

Data	long	->	wide
Number of obs.	30	->	6
Number of variables	5	->	12
j variable (5 values)	trialnum	->	(dropped)
xij variables:			
	bp	->	bp1 bp2 ... bp5
	pl	->	pl1 pl2 ... pl5

Contains data

```

obs:          6
vars:         12

```

```
size:          132
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
bp1	int	%3.0f	1 bp	
pl1	int	%3.0f	1 pl	
bp2	int	%3.0f	2 bp	
pl2	int	%3.0f	2 pl	
bp3	int	%3.0f	3 bp	
pl3	int	%3.0f	3 pl	
bp4	int	%3.0f	4 bp	
pl4	int	%3.0f	4 pl	
bp5	int	%3.0f	5 bp	
pl5	int	%3.0f	5 pl	
age	byte	%3.0f		Age of person

```
Sorted by: id
```

```
Note: Dataset has changed since last saved.
```

	id	bp1	pl1	bp2	pl2	bp3	pl3	bp4	pl4	bp5	pl5	age
1.	1	115	54	86	87	129	93	105	81	127	92	40
2.	2	123	92	136	88	107	125	111	87	120	58	30
3.	3	124	105	122	97	101	128	109	57	112	68	16
4.	4	105	52	115	79	121	71	129	106	137	39	23
5.	5	116	70	128	64	112	52	125	68	111	59	18
6.	6	108	74	126	78	124	92	131	99	107	80	27

4.5 Reshaping wide to long problems

Problem 1: Failing to list all of the varying variables to be reshaped.

This is a dangerous kind of problem, since it doesn't throw an error. This requires the user to inspect the data after reshaping to make sure everything worked properly.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide, clear
reshape long bp, i(id) j(trialnum)
list
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

(note: j = 1 2 3 4 5)

Data	wide	->	long
Number of obs.	6	->	30
Number of variables	12	->	9
j variable (5 values)		->	trialnum
xij variables:	bp1 bp2 ... bp5	->	bp

	id	trialnum	age	bp	p11	p12	p13	p14	p15
1.	1	1	40	115	54	87	93	81	92
2.	1	2	40	86	54	87	93	81	92
3.	1	3	40	129	54	87	93	81	92
4.	1	4	40	105	54	87	93	81	92
5.	1	5	40	127	54	87	93	81	92
6.	2	1	30	123	92	88	125	87	58
7.	2	2	30	136	92	88	125	87	58
8.	2	3	30	107	92	88	125	87	58
9.	2	4	30	111	92	88	125	87	58
10.	2	5	30	120	92	88	125	87	58
11.	3	1	16	124	105	97	128	57	68
12.	3	2	16	122	105	97	128	57	68
13.	3	3	16	101	105	97	128	57	68
14.	3	4	16	109	105	97	128	57	68
15.	3	5	16	112	105	97	128	57	68
16.	4	1	23	105	52	79	71	106	39
17.	4	2	23	115	52	79	71	106	39
18.	4	3	23	121	52	79	71	106	39
19.	4	4	23	129	52	79	71	106	39
20.	4	5	23	137	52	79	71	106	39

21.	5	1	18	116	70	64	52	68	59
22.	5	2	18	128	70	64	52	68	59
23.	5	3	18	112	70	64	52	68	59
24.	5	4	18	125	70	64	52	68	59
25.	5	5	18	111	70	64	52	68	59
26.	6	1	27	108	74	78	92	99	80
27.	6	2	27	126	74	78	92	99	80
28.	6	3	27	124	74	78	92	99	80
29.	6	4	27	131	74	78	92	99	80
30.	6	5	27	107	74	78	92	99	80

The remedy is easy: don't forget all of your variables to be reshaped

```
use cardio_wide, clear
reshape long bp pl, i(id) j(trialnum)
```

Problem 2: Be careful when the time-varying value is embedded within the variable name and not at the end.

Typically you will find that the varying variable will have $var_1, var_2, \dots, var_k$ and you simply specify `reshape long var, i(id) j(time)`. But, if the varying value is embedded like `t1bp, ..., t5bp` and `t1pl, ..., t5pl`, then we cannot simply use `reshape long tbp tpl, i(id) j(trialnum)`, since Stata looks for the varying value at the end.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide2, clear
describe
capture reshape long tpb, i(id) j(trialnum)
reshape error
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

Contains data from cardio_wide2.dta

```
obs:      6
vars:     12
size:     120
```

31 Dec 2009 15:46

variable name	storage type	display format	value label	variable label
---------------	--------------	----------------	-------------	----------------

Sorted by:

```
reshape looked for existing variables named a# and b# but could not find any. Rem
```

```
long to wide: reshape wide a b, i(i) j(j)    (j existing variable)
wide to long: reshape long a b, i(i) j(j)    (j new variable)
```

Data wide -> long


```

Number of obs.                6   ->    30
Number of variables            12  ->     5
j variable (5 values)         ->  trialnum
xij variables:
      t1p1 t2p1 ... t5p1  ->  t1p
      t1bp t2bp ... t5bp  ->  t1bp

```

Contains data

```

obs:      30
vars:      5
size:     210

```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
trialnum	byte	%9.0g		
age	byte	%3.0f		Age of person
tbp	int	%3.0f		
tpl	int	%3.0f		

Sorted by: id trialnum

Note: Dataset has changed since last saved.

```

+-----+
| id  trialnum  age  tbp  tpl |
+-----+
1. | 1      1    40  115  54 |
2. | 1      2    40   86  87 |
3. | 1      3    40  129  93 |
4. | 1      4    40  105  81 |
5. | 1      5    40  127  92 |
+-----+
6. | 2      1    30  123  92 |
7. | 2      2    30  136  88 |
8. | 2      3    30  107 125 |
9. | 2      4    30  111  87 |
10. | 2      5    30  120  58 |
+-----+

```

Problem 3: Be specific with which variables to reshape and don't mistakenly constant variables that have the same prefix as the varying variables

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide3, clear
clist bp* pl*, noobs
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

bp1	bp2	bp3	bp4	bp5	bp2005	pl1	pl2	pl3	pl4	pl5	pl2005
115	86	129	105	127	112	54	87	93	81	92	81
123	136	107	111	120	119	92	88	125	87	58	90
124	122	101	109	112	113	105	97	128	57	68	91
105	115	121	129	137	121	52	79	71	106	39	69
116	128	112	125	111	118	70	64	52	68	59	62
108	126	124	131	107	119	74	78	92	99	80	84

Our *bp2005* and *pl2005* are constant variables within our cross-section, but have the same prefix as our value-varying variables.

```
reshape long bp pl, i(id) j(trialnum)
describe
list in 1/12, sepby(id)
```

(note: j = 1 2 3 4 5 2005)

Data	wide	->	long
Number of obs.	6	->	36
Number of variables	14	->	5
j variable (6 values)		->	trialnum
xij variables:			
	bp1 bp2 ... bp2005	->	bp
	pl1 pl2 ... pl2005	->	pl

Contains data

```
obs:      36
vars:      5
```

```
size:          432
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
trialnum	int	%9.0g		
age	byte	%3.0f		Age of person
bp	float	%9.0g		
pl	float	%9.0g		

Sorted by: id trialnum

Note: Dataset has changed since last saved.

```
+-----+
| id  trialnum  age  bp  pl |
+-----+
1. | 1      1    40  115  54 |
2. | 1      2    40   86  87 |
3. | 1      3    40  129  93 |
4. | 1      4    40  105  81 |
5. | 1      5    40  127  92 |
6. | 1    2005    40  112  81 |
+-----+
7. | 2      1    30  123  92 |
8. | 2      2    30  136  88 |
9. | 2      3    30  107 125 |
10. | 2      4    30  111  87 |
11. | 2      5    30  120  58 |
12. | 2    2005    30  119  90 |
+-----+
```

We did not intend to reshape *bp2005* and *pl2005*.

To remedy, we need to specify the values within `j()` with `j(trialnum 1-5)` to indicate that we only want to reshape suffixes with 1-5.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide3, clear
reshape long bp pl, i(id) j(trialnum 1-5)
describe
list in 1/10, sepby(id)
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```

Data                                wide  ->  long
-----
Number of obs.                      6  ->    30
Number of variables                  14  ->     7
j variable (5 values)                ->  trialnum
xij variables:
      bp1 bp2 ... bp5  ->   bp
      pl1 pl2 ... pl5  ->   pl
-----

```

```

Contains data
  obs:      30
  vars:      7
  size:     450
-----

```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
trialnum	byte	%9.0g		
age	byte	%3.0f		Age of person
bp	int	%3.0f		
bp2005	float	%9.0g		
pl	int	%3.0f		
pl2005	float	%9.0g		

Sorted by: id trialnum

Note: Dataset has changed since last saved.

	id	trialnum	age	bp	bp2005	pl	pl2005
1.	1	1	40	115	112	54	81
2.	1	2	40	86	112	87	81
3.	1	3	40	129	112	93	81
4.	1	4	40	105	112	81	81
5.	1	5	40	127	112	92	81
6.	2	1	30	123	119	92	90
7.	2	2	30	136	119	88	90
8.	2	3	30	107	119	125	90
9.	2	4	30	111	119	87	90
10.	2	5	30	120	119	58	90

+-----+

Problem 4: No id variable to identify the cross-sectional unit

If each wide-format observation is a unique cross-sectional unit, then we can use `_n` to generate a unique id variable.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide3, clear
describe
gen idcode = _n
describe
reshape long bp pl, i(id idcode) j(trialnum 1-5)
describe
list in 1/10, sepby(id)
```

/Users/Sam/Desktop/Econ 645/Data/Mitchell

Contains data from cardio_wide3.dta

```
obs:      6
vars:     14
size:     168
31 Dec 2009 15:46
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
age	byte	%3.0f		Age of person
bp1	int	%3.0f		Blood pressure systolic Trial 1
bp2	int	%3.0f		Blood pressure systolic Trial 2
bp3	int	%3.0f		Blood pressure systolic Trial 3
bp4	int	%3.0f		Blood pressure systolic Trial 4
bp5	int	%3.0f		Blood pressure systolic Trial 5
bp2005	float	%9.0g		
pl1	int	%3.0f		Pulse: Trial 1
pl2	byte	%3.0f		Pulse: Trial 2
pl3	int	%3.0f		Pulse: Trial 3
pl4	int	%3.0f		Pulse: Trial 4
pl5	byte	%3.0f		Pulse: Trial 5
pl2005	float	%9.0g		

Sorted by:

Contains data from cardio_wide3.dta

```
obs:      6
vars:     15
size:     216
```

31 Dec 2009 15:46

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
age	byte	%3.0f		Age of person
bp1	int	%3.0f		Blood pressure systolic Trial 1
bp2	int	%3.0f		Blood pressure systolic Trial 2
bp3	int	%3.0f		Blood pressure systolic Trial 3
bp4	int	%3.0f		Blood pressure systolic Trial 4
bp5	int	%3.0f		Blood pressure systolic Trial 5
bp2005	float	%9.0g		
pl1	int	%3.0f		Pulse: Trial 1
pl2	byte	%3.0f		Pulse: Trial 2
pl3	int	%3.0f		Pulse: Trial 3
pl4	int	%3.0f		Pulse: Trial 4
pl5	byte	%3.0f		Pulse: Trial 5
pl2005	float	%9.0g		
idcode	double	%10.0g		

Sorted by:

Note: Dataset has changed since last saved.

Data	wide	->	long
Number of obs.	6	->	30
Number of variables	15	->	8
j variable (5 values)		->	trialnum
xij variables:			
	bp1 bp2 ... bp5	->	bp
	pl1 pl2 ... pl5	->	pl

Contains data

```
obs:      30
vars:      8
size:     690
```

variable name	storage type	display format	value label	variable label
id	byte	%3.0f		ID of person
idcode	double	%10.0g		
trialnum	byte	%9.0g		
age	byte	%3.0f		Age of person
bp	int	%3.0f		
bp2005	float	%9.0g		
pl	int	%3.0f		
pl2005	float	%9.0g		

Sorted by: id idcode trialnum

Note: Dataset has changed since last saved.

	id	idcode	trialnum	age	bp	bp2005	pl	pl2005
1.	1	1	1	40	115	112	54	81
2.	1	1	2	40	86	112	87	81
3.	1	1	3	40	129	112	93	81
4.	1	1	4	40	105	112	81	81
5.	1	1	5	40	127	112	92	81
6.	2	2	1	30	123	119	92	90
7.	2	2	2	30	136	119	88	90
8.	2	2	3	30	107	119	125	90
9.	2	2	4	30	111	119	87	90
10.	2	2	5	30	120	119	58	90

4.6 Multilevel datasets

Mitchell discusses when we have multiple levels of data, such as cross-sectional unit i and time dimension t , where data at level 1 are constant across time within unit, but vary across cross-sectional units and level 2 data are time-varying data that vary within a cross-sectional unit.

Sometimes we have multiple levels, such as school district, school, classroom, and student.

If you are interested, please review pages 311-314

4.7 Collapsing datasets

The *collapse* command can be useful finding statistics at aggregate levels for different groups. You don't have to use *egen* but you can if you would like. After using the *egen* command, such as *egen sumvar = sum(var)*, or *egen avgvar = mean(var)* the *collapse* command can be useful.

Let's say we want to aggregate mean wages by state in the CPS, we can use the *egen mean* command to find average wages in the state for month *m* in year *y*. We can then keep state month year and *avgvar* for all 50 states for each month and year of interest.

We need to be careful with the *collapse* command, since your original data are lost from memory unless you use a tempfile or a data frame.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
list, sepby(id)
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

	id	trial	age	bp	pl
1.	1	1	40	115	54
2.	1	2	40	86	87
3.	1	3	40	129	93
4.	1	4	40	105	81
5.	1	5	40	127	92
6.	2	1	30	123	92
7.	2	2	30	136	88
8.	2	3	30	107	125
9.	2	4	30	111	87
10.	2	5	30	120	58
11.	3	1	16	124	105
12.	3	2	16	122	97
13.	3	3	16	101	128
14.	3	4	16	109	57
15.	3	5	16	112	68
16.	4	1	23	105	52
17.	4	2	23	115	79
18.	4	3	23	121	71
19.	4	4	23	129	106

20.		4		5		23		137		39	

21.		5		1		18		116		70	
22.		5		2		18		128		64	
23.		5		3		18		112		52	
24.		5		4		18		125		68	
25.		5		5		18		111		59	

26.		6		1		27		108		74	
27.		6		2		27		126		78	
28.		6		3		27		124		92	
29.		6		4		27		131		99	
30.		6		5		27		107		80	

		+									

Let's say we want to find mean blood pressure and pulse for each cross-sectional unit, then we can use the egen command or the collapse commands.

```
sort id
by id: egen meanbp = mean(bp)
by id: egen meanpl = mean(pl)
by id: egen maxbp = max(bp)
by id: egen maxpl = max(pl)
by id: egen minbp = min(bp)
by id: egen minpl = min(pl)
by id: egen medbp = pctlbp(bp), p(50)
by id: egen medpl = pctlbp(pl), p(50)

collapse meanbp meanpl maxbp maxpl minbp minpl medbp medpl, by(id)
```

Or you can just use the collapse command

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear

collapse (mean) meanbp=bp meanpl=pl (max) maxbp=bp maxpl=pl (min) minbp=bp minpl=pl (p50) medbp=b
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

Notice that our data are rounded to the nearest 1, unlike our egen commands, we can format our data to reflect that decimal point to two places instead of rounding up by default

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
format bp* pl* %5.2f
collapse (mean) meanbp=bp meanpl=pl (max) maxbp=bp maxpl=pl (min) minbp=bp minpl=pl (p

/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

4.8 Exercise

Pull BLS monthly inflation data, reshape it into long, and merge with CPS

Long, J.S, and Freese, J. (2006) *Regression Models for Categorical Dependent Variables Using Stata* (2nd ed.). Stata Press.

Mitchell, N.M. (2020). *Data Management Using Stata: A Practical Handbook*. (2nd ed.). Stata Press.

UCLA Advanced Research Computing. (2025). *How can I access information stored after I run a command in Stata*. Retrieved from <https://stats.oarc.ucla.edu/stata/faq/how-can-i-access-information-stored-after-i-run-a-command-in-stata-returned-results/>

Wooldridge (2009). *Introductory Econometrics: A Modern Approach*. (4th ed.). South-Western Cengage Learning.